



Research Paper

The wheel of life? The effect of the abolition of the foundling wheel in nineteenth-century Italy

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ABSTRACT

This paper examines the effects of abolishing the foundling wheel (*ruota*) on reproductive decision-making in post-unitary Italy (1863–1882). The *ruota* was a turning wheel placed on a wall outside foundling homes across Catholic Europe, which offered a means for anonymous infant abandonment. As infant abandonment rates and foundling mortality soared in the nineteenth century, countries began dismantling these systems. Italy mirrored this trend, with provinces abolishing the *ruota* at different times. We investigate the specific impacts of the *ruota* abolition on infant abandonment, infant mortality, births, and gender discrimination. To do so, we use a novel longitudinal dataset of Italian provinces and a staggered Difference-in-Difference empirical design. Our findings show a 54.9% decrease in abandonment, a 10.4% decline in infant deaths, and a 4% drop in births after the abolishment of the *ruota*. This suggests a significant shift in reproductive behavior and family planning in response to policy change.

“I have seen Garibaldi crying only twice.

The first one was when we visited the foundlings’ home in Palermo.

The second one when we visited the foundlings’ home in Naples”

[Jessie White Mario (1897) Writer, philanthropist, and Garibaldi’s nurse in four wars]

1. Introduction

A rational approach to reproductive decision-making that entails some control over new birth planning is an issue relevant for both micro (individual) and macro (policy) economic decisions (Becker, 1993; Bloom et al., 2003; Galor and Weil, 1999, 2000). This is a classic economic problem: limits to population growth are crucial to escape the Malthusian trap. In the past, tools to apply this judgmental control were limited. Before the diffusion of scientific contraception in the late nineteenth century, abstinence and withdrawal were the most common measures adopted (Guinnane, 2011). When these failed, precarious abortions, infanticides (violent or by neglect), and abandonment were the outcome (Langer, 1972; Sauer, 1978; Hanlon, 2016, 2023). From the second half of the eighteenth century, infant abandonment had reached alarming levels across Europe, with devastating consequences for foundlings’ mortality rates and public finances because of the publicly funded provision of care. In Italy alone, a staggering

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number of more than 600,000 children were abandoned in 1863–1882, ca. 3% of all newborns in the period. Italy does not seem an extreme case in a European perspective (Derosas and Tsuya, 2010). In the mid-nineteenth century, foundling ratios per 100 births were notably high in various European capitals. Paris witnessed rates between 20 and 35 (Meyer, 1980), Madrid and Dublin around 20, Prague 30, and Vienna a striking 50 (Kertzer, 1993).¹ The fate of foundlings, however, represents a gloomy page of European history. Mortality rates among abandoned children were much higher than in the general population (Kok et al., 1997; Levene, 2005). Overall, in the period 1866–1875, over 152,050 Italian children were admitted to foundlings homes. Of these, 58,827, i.e. approximately 39% of the total admissions, died before reaching the age of one (MAIC, 1884).

Since the eighteenth century, institutions had been developed to deal with the uncontrolled increase in child abandonment. Examples include the establishment of facilities where children could be legally and anonymously abandoned, with the aim to preserve families, and in particular women, from public disgrace and scandal (Hunecke, 1991; Tilly et al., 1992).² One of these institutions, found especially in Southern European and Catholic countries, was the *ruota* (foundling wheel): a rotating cylinder embedded in the wall of foundling hospitals or churches, allowing people to place an infant inside from the street and rotate it into the building, effectively leaving the child anonymously (Boswell, 1998). After several attempts to discourage abandonment, in the nineteenth century countries began phasing out foundling wheels and replacing them with consignment offices. At these offices, those who wished to abandon an infant were required to disclose their identities. Belgium and France began to phase out the *ruote* after 1840 (Potash, 1979), followed by Spain in the 1850s, and Portugal in the 1860s (Ransel, 1988). Similarly, in Italy, following a heated public debate (Gorni and Pellegrini, 1974), the device fell into general disrepute. Ferrara, in Emilia–Romagna, was the first province to shut down the *ruota* in 1867, and ten years later, the number of foundling wheels in Italy had halved.

This paper studies the effects of the abolition of the *ruota* in nineteenth-century Europe, focusing on post-unitary Italy (1863–1882) as a case study. Thus, our research question is: What were the effects of the abolition of the *ruota* on reproductive decision-making and children's survival chances in post-unitary Italy? Specifically, we study the impact of the abolition of the *ruota* on infant abandonment, births, infant mortality, and various dimensions of gender discrimination. Nineteenth-century Italy represents the ideal setting to conduct an event study on the effect of the closure of the *ruote*, as we can exploit the fact that they were closed in different years across Italian provinces.

To do so, we construct a novel longitudinal dataset at the provincial level for the period 1863–1882. The main data source was the *Movimento dello Stato Civile*, a government-published annual statistical report assembled by the Directorate of Statistics which provides detailed data on demographic data collected through the Civil Registry (*Registro dello Stato Civile*), such as the number of births (legitimate and illegitimate), of foundlings, marriages, and deaths by age group.³ All these variables are disaggregated by gender. Our identification strategy relies on a Staggered Difference-in-Difference estimation (Callaway and Sant'Anna, 2021). Our treatment is the abolition of the *ruota*, which happened with a staggered timing across provinces. This estimation method allows us to deal with the heterogeneity not only in the timing, but also in the dissemination of the treatment.

We find that abolishing the *ruota* decreased the number of total abandonment by 54.2%. We find a negative and significant effect on the number of births of 4.1%. To shed light on the potential mechanisms, we estimate the effect of closing *ruota* on legitimate and illegitimate births separately, and we find that the decline in fertility was driven mainly by marital fertility. We interpret these results as a behavioral change that also influenced reproductive control and family planning strategies. Furthermore, the effect of the abolition of the *ruota* on children's mortality is negative and significant (thus, the effect on children survival is positive). The closure led to an overall decline in infant deaths of 10.1% — a result consistent with improved care and more resources dedicated to children. Last, we find that closing the *ruota* reduced the sex selectivity in abandonment, which had been historically biased towards women.

This study holds significant interest for economic history, historical demography, and economics literature. First, it contributes to the social history of child abandonment by analyzing the effect of a very important policy: the abolition of the foundling wheel. Historical literature has identified the pivotal role of the foundling wheel in reducing the number of admissions to foundling homes (Corsini, 1987; Viazso et al., 1997). The present study advances this understanding by contending that the closures represented a critical turning point in the social history of abandonment, not only because they led to a substantial decline in the number of foundlings, thus altering the composition of the founding population, but also because they affected the survival chances of infants, both within and outside foundling homes, and fertility decisions. In 1993, Kertzer used qualitative archival evidence to demonstrate how pervasive the policy was in shaping families' reproductive behavior. The present study aims to provide quantitative evidence of this phenomenon. In addition, we analyze a specific dimension of child abandonment, namely sex selectivity. Beltrán Tapia and Cappelli (2024) have provided further insights into this topic within the broader context of the determinants of child abandonment in Europe (Beltrán Tapia and Szoltysek, 2022). They hypothesize that sex-selective abandonment contributed to the unexplained excess female mortality they find. This paper further explores this relationship by examining the effect of closures on the sex composition of abandonment and on the sex ratios in infant population and mortality.

Second, this paper contributes to our understanding of how institutional changes around child abandonment affected fertility decline in Italy, with broader implications for the quantity–quality trade-off theory. Prior work has suggested the Italian fertility transition began in the late nineteenth century and accelerated in the early twentieth century (Livi Bacci, 1977, 1986). However,

¹ These numbers should be taken critically because it is likely that many newborns in the immediate countryside were transported and abandoned in the capital city, thus inflating these figures.

² Abandonment has existed since antiquity, but it is only with modern states that we witness a transition from a system based on the “kindness of strangers” to a more institutionalized one (Boswell, 1998).

³ Legitimate children were born within the wedlock, whereas illegitimate children were not.

more recent works (Guinnane, 2011) have challenged the reliability of the demographic data and analytical methods used in these earlier studies. They suggest a re-thinking of some of the conclusions of the Princeton Project on the Decline of Fertility in Europe (EPF), in particular concerning the minor role it attributes to social, cultural, and economic factors in shaping the timing and pace of the fertility transition. Economic analyses of fertility transition, also applied to historical contexts, are based on the microeconomic models of fertility developed by Gary Becker, one of the founders of the New Home Economics (NHE). Since at least the 1960s, family planning and its intertwining with economic incentives and outcomes have featured prominently in the NHE literature, which develops the conceptual framework of a quantity–quality (Q–Q) model of fertility decisions (Becker, 1960, 1993; Bailey et al., 2014; Olsen, 1994; Cleland et al., 2006; Canning and Schultz, 2012; Bloom et al., 2003; Karra and Wilde, 2023; Doepke et al., 2023; Alam and Pörtner, 2018; Levine, 2020).⁴ The rationale is that children are not an inferior good, and as couples become richer, for example after industrialization, the opportunity cost of having children increases. Consequently, they invest more in the quality of children and reduce their number. Guinnane (2011) criticizes Beckers’ Q–Q framework concerning the increase in the direct cost of children as an incentive for couples to have smaller families, since he argues that most costs did not change in the relevant period in the way we would expect a fertility decline to have done. A similar dynamic has been observed in the United States, albeit in a different historical and institutional context. Aaronson et al. (2014) analyze the effects of a large-scale school construction program in the American South and its impact on the fertility decline of rural Black communities between 1910 and 1930. Their findings suggest that policies increasing the cost of child-rearing, even when not explicitly designed for demographic purposes, can lead to significant reductions in fertility rates.⁵ Starting from the 1990s, a further aspect of NHE’s predictions has been severely criticized by feminist economics: the rational decision-making of the fertility decision. Feminist economics rejects the concept of *homo oeconomicus* and recognizes the historical subordination and discrimination of women’s work, focusing on the importance of care, power dynamics, cultural norms, and interdependence between individuals (Becchio, 2019; Ferber and Nelson, 1993; Folbre, 2006).⁶ Our paper contributes to this debate in several ways: First we suggest that anonymous abandonment guaranteed by the *ruota* was one of the most common preventive checks available in preindustrial Italy.⁷ Second, its closure led to an increase in the cost of child-rearing before the country’s industrialization and before what has been commonly identified as the start of the fertility transition. This increase in cost happened during the relevant period and had an impact on infant mortality and family reproductive strategies. It was not the consequence of an economic explanation, as industrialization or urbanization. In fact, the *ruota* was a path-dependent institution profoundly embedded in the social and cultural milieu: its existence and abolition were linked to changes in cultural norms. Our findings can be interpreted using the framework “Ready, Willing, Able” (RWA), first developed by Coale (1973) and used in the demographic literature.⁸ Hacker (2016) applies this framework to the analysis of the timing of marital fertility in the second half of the 19th century in the United States and finds that economic, cultural, and legal obstacles were confounding factors in delaying the decline in marital fertility. Moreover, he argues that the literature has predominately focused on economic incentives, so on the readiness, and advocates for the use of more inclusive models of fertility. The decrease in cost, which is a consequence of abolition, can be interpreted as leading to a change in couples’ willingness to control fertility.

The paper proceeds as follows. Section 2 provides the historical context of the abandonment system in Italy and summarizes the debate over the closure of the *ruota*. Section 3 describes the sources. Section 4 the methodology. Section 5 reports the main results. Section 6 concludes.

2. Historical background

2.1. Infant abandonment in European perspective

Infant abandonment was a massive phenomenon in Europe from the end of the Middle Ages to the nineteenth-century. In particular, since the Reformation, it became a distinguishing feature of Catholic Europe, and made only modest inroads in Protestant countries (Fuchs, 1987; Ransel, 1988; Kertzer, 1993).⁹ The abandonment process was quite similar throughout the continent.¹⁰ century China and Japan often resorted to prenatal family planning, and in particular to infanticide (Derosas and Tsuya, 2010).

⁴ For instance Willis (1999) incorporates children born out-of-wedlock into Becker (1960)’s model of marriage and fertility. This attempt is particularly relevant in the context of this paper.

⁵ Unlike our study, which focuses on a specific institutional change affecting the costs of child-rearing, their research examines both intensive and extensive fertility margins, including the decision to remain childless.

⁶ Feminist economics has criticized multiple dimensions of the NHE which are relevant for the present work. Gender power dynamics have been culpably neglected by models applying economic rationality to human behavior (England, 1993; Folbre, 2001), as well as women’s unpaid and domestic work (Folbre, 1994; Nelson, 1995). Moreover, individuals are treated as isolated entities by the NHE. Thus, according to feminist economists, it has failed to account both for the interdependency of human relations (Nelson, 2006; Folbre and England, 1999) and for the importance of institutional contexts, such as policies, cultural norms, and institutional support.

⁷ Cinnirella et al. (2017, 2019) argue that in England even before 1800 some fertility control was attained using rudimentary contraceptive practices, resulting into birth spacing among poor families. However, their findings have been criticized by Clark and Cummins (2019) who do not find any evidence of parity-dependent fertility — i.e., fertility choices depending on the number of surviving offspring of the family — in pre-1850 England.

⁸ *Ready* has to do with economic incentives that induce couples to perceive smaller families as economically advantageous. *Willing* has to do with changing norms, in a way that make the practice of fertility control ethically acceptable. *Able* is about the availability of fertility controls. See Lesthaeghe and Vanderhoeft (2001), Hacker (2016) for a more detailed description of the RWA framework.

⁹ For instance, London founding hospital was opened only in 1739, three centuries later the Florentine famous *Ospedale degli Innocenti* (Berry, 2019). An exception is Orthodox Russia, which also faced staggering abandonment and, starting from the eighteenth century, adopted care facilities for foundlings following the French system as a model (Ransel, 1988).

¹⁰ Eastern and Western pre-industrial societies employed different strategies to control family size. In historical Europe, infant abandonment (postnatal control) was a common practice. In contrast, eighteenth- and nineteenth

Most abandoned babies were left when they were only a few days old and put in a rotating cradle (in Italian, *ruota*) placed on a wall outside foundling homes or churches, where parents could abandon them while maintaining anonymity. Abandoned babies would spend only a few days in the foundling home and were then sent to wetnurses, women employed by foundling institutions usually living in the countryside, for fostering and breastfeeding.¹¹ At the time of their establishment in the Italian peninsula, foundling homes took in children regardless of their legal status (i.e., legitimate vis-à-vis illegitimate). It was not until the eighteenth century that they began to enact laws to prevent the abandonment of legitimate infants. However, as long as the *ruota* existed and anonymity was guaranteed, legitimate foundlings could not really be excluded. The push to exclude legitimate children from public assistance coincided with the aspiration to transform foundling institutions into places to host *all* illegitimate children, since unwed mothers were deemed unfit to raise their own children and compelled to abandon them to preserve their honor (Povolo, 1989).¹² In Catholic Europe, across centuries, mechanisms ensuring the unwed mothers' anonymity were refined through extensive networks involving local communities, neighbors, priests, and even midwives obligated to report pregnancies to authorities (Kertzer, 1993).

In the period preceding the abolition of the *ruota*, the proportion of legitimate births among the total number of abandoned infants exhibited variation across different cities.¹³ Hunecke (1991) states that in the nineteenth century, the percentage of legitimate births in Milan's foundling home was approximately 50–70, while in Brescia, Cremona and Florence, it was between 40 and 70. By analyzing the baptism certificates of foundlings, Corsini (1976) was able to reconstruct the family history of a small sample of foundlings originating from the historic province of foundlings. His findings indicate that children from families who only abandoned one child were predominantly of lower birth order. In contrast, those born into families who abandoned more than one child were of higher birth order. Understanding how the foundling population was stratified in terms of legitimacy status and birth order and its evolution over time permits to understand better our findings on the reduction of births, by disentangling whether it was prominently led by a decline in marital or non-marital fertility.

The differentiation by gender is a further important dimension of the foundling population. Selective abandonment has historically disadvantaged girls. In the earlier centuries, when mostly married couples resorted to abandonment, the preference for abandoning girls over boys was marked. In the period of analysis, sex selectivity was less marked as unwed parents were forced to abandon their child, regardless of the child's sex. In conclusion, the decline from the Middle Ages to the nineteenth century in sex selectivity went hand in hand with changes in the extent to which illegitimate abandonment prevailed on legitimate (Beltrán Tapia and Cappelli, 2024).

2.2. The *ruota* in nineteenth-century Italy

Foundling institutions (in Italian, *brefotrofi*) existed throughout the Italian peninsula in the nineteenth century, but some of their practices varied across regions and even within pre-unitary states. As mentioned above, the most common method of accepting abandoned infants was the *ruota*. The distribution and size of these *ruote* differed significantly.

In some provinces, such as Milan and Florence, a single large institution with a *ruota* served the entire area. The *Pia Casa degli esposti e delle partorienti in Santa Caterina alla ruota* of the *Ospedale Maggiore* in Milan, and the *Ospedale degli Innocenti* in Florence were among the oldest and largest foundling hospitals in Europe, reflecting the long history of organized child abandonment in those cities. In other regions, particularly the South, many smaller local *ruote* were scattered across towns and villages. For example, in Sicily, *ruote* could be found in over 350 municipalities as late as 1890. The left-hand side map in Fig. 1 shows this heterogeneity in the number of *ruote* by provinces in the aftermath of the country's Unification in 1861.¹⁴ Unification, Italy was divided into 69 provinces. These provinces were based on pre-unification territorial divisions of the Italian states and were subject to slight modifications as territories were annexed (such as the inclusion of Rome in 1870). The number of municipalities was several orders of magnitude greater than the number of provinces. By the 1871 census, Italy had approximately 8259 municipalities.

The presence and resources of foundling institutions were also influenced by the boundaries of pre-unitary states and the uneven distribution of charitable organizations. The Kingdom of Sardinia in the North and the Grand Duchy of Tuscany in the Center had well-established networks of hospitals and orphanages, often with significant endowments from the Catholic Church or wealthy benefactors. In contrast, the Kingdom of the Two Sicilies in the South lacked a centralized system of poor relief, relying instead on local initiatives and religious orders.

These regional disparities persisted after Italian unification in 1861, as the new national government struggled to harmonize the various legal and administrative frameworks governing child welfare. The Casati Law of 1859, which had modernized the founding system in Piedmont by introducing stricter admission criteria and promoting foster care, was not immediately extended to other regions. Instead, the Italian Parliament debated several proposals for a unified national law on foundlings in the 1870s and 1880s, but none were ultimately approved.

¹¹ However, when the numbers were too large, not enough wetnurses were available, and babies stayed in the foundlings hospital. There, they were fostered with nursing bottles and by the few in-house wetnurses. In these overcrowded conditions, their chances of surviving were slimmer (Viazzo et al., 1997).

¹² According to White Mario (1897), the anonymity of the *ruota*, together with the impossibility to search for fatherhood sanctioned by the Civil Code, allowed the Catholic clergy to hide with impunity their breaking of the celibacy.

¹³ In the presence of the *ruota*, it was difficult to quantify foundlings by legitimacy status. To make an estimate, historians and historical demographers have used the information contained in the letters or the tokens sometimes left by parents when they abandoned their child. Sometimes, abandoned children were left with their baptism certificates, thus making it possible for scholars to reconstruct their family history. Trials for the return of children mostly concerned legitimate foundlings. They are also a useful source to obtain information on family structure.

¹⁴ In Italy, a *provincia* (province) is a larger administrative division, encompassing several *comuni* (municipalities). Each municipality is a local government unit with its own mayor and council, while the province coordinate services across municipalities. After the

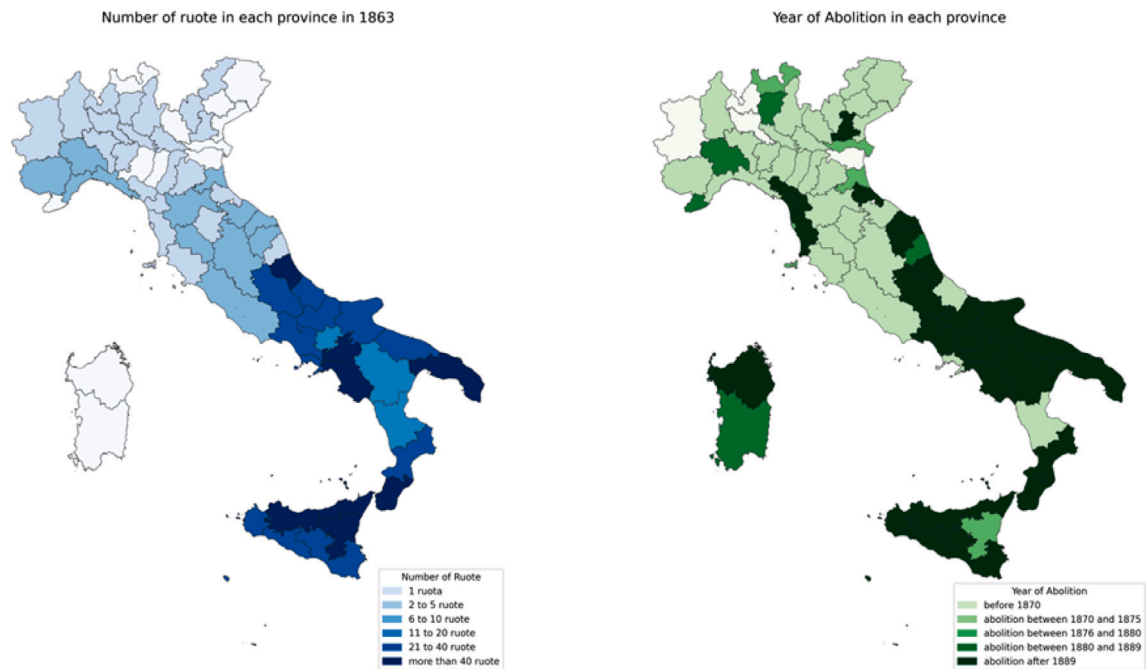


Fig. 1. Number of *ruote* in 1863 and year of abolition across Italian provinces.

2.3. The abolition of the *ruota*

Before 1866, a *ruota* existed in all provinces of Italy, for a total of 1179 municipalities. In 1867, Ferrara was the first province to close its *ruota*, followed by Milan the next year. Gradually, a growing number of provincial Councils approved their abolition, replacing them with consignment offices. By 1880, the *ruota* still existed only in 675 municipalities, and this number dropped to 306 in 1897.¹⁵ The right-hand side map in Fig. 1 shows the years of abolition across Italian provinces, which follows clear geographical patterns.

The abolition movement was triggered by the Law of 20 March 1865, which reorganized the competencies between the newborn Italian State and the peripheral institutions, such as provinces and municipalities. Before 1859, the organization of foundling institutions was regulated by local laws and was often under the (financial) responsibility of the pre-unitary States aided by municipalities and local charity institutions — if present (*dei Deputati*, 1874). By contrast, the 1865 law put all the financial burden on municipalities and provinces. The abolition of the *ruota* was up to the provincial Councils.¹⁶ In fact, the 1865 law merely included a transitory measure that made the provincial Councils and municipalities responsible for the expenses incurred by local foundling hospitals until a proper national law would thoroughly regulate the matter. The new municipal and provincial law (approved with a Royal Decree on 4 May 1898) simply renewed the transitory measures. An all-embracing national law regulating the issue of foundlings and their institutions was enacted only in 1923. This implied that a large degree of autonomy was left to provincial Councils in dealing with foundlings throughout the period.

The rationale for closing the *ruota*, and thus halting anonymous children abandoning, emerges clearly from coeval documents such as minutes of provincial Council meetings, Law proposals, and reports from public officials and medics.¹⁷ Both cultural and economic concerns played a role. As discussed by Gorni and Pellegrini (1974), the abolition of the *ruote* followed a heated public debate. Some contemporary observers saw its closure as the only way to make abandonment and the staggering mortality rates in the overcrowded foundling homes decline.¹⁸ Defenders of the *ruota* justified its existence as the only viable alternative to infanticides and abortions, as they feared that these would have skyrocketed if anonymity was repealed.¹⁹ They also feared that, absent the

¹⁵ All remaining *ruote* were closed in 1923 when the new *Regolamento generale per il servizio d'assistenza agli Esposti* was enacted by the first Mussolini government.

¹⁶ The expenses were distributed differently across Italian regions. In Piedmont, municipalities covered one-quarter of the expenses, while provinces only three-quarters. In Tuscany, Marche, and Umbria, municipalities bore two-thirds of the costs, and the province covered one-third. In Emilia and most provinces in the former Kingdom of two Sicilies, the expenses were equally divided between municipalities and the province. In Lombardy and Veneto, the provincial budgets nearly fully covered the costs. Remarkably, the province of Milan completely exempted municipalities from these expenses (Rasari, 1881).

¹⁷ Bertolini (1893), Buffini (1840), *dei Deputati* (1874), De Crescenzo (1873), De Vincenti (1861), Grillenzoni (1861), Nicotera (1878), Rasari (1881), Scotti (1877), Tocci (1878).

Table 1

Provincial expenses 1871–1884.

Source: MAIC (1886), Bilanci Provinciali per gli anni 1883–1884, Roma: Tipografia Romana.

Year	Charity expenses	Foundling expenses	Total expenses	Charity %	Esposti %
1871	15,274,292	8,063,022	80,370,006	19%	10%
1872	15,589,230	8,060,437	87,933,639	18%	9%
1873	14,230,179	6,956,514	79,473,050	18%	9%
1874	15,093,631	7,015,970	82,191,600	18%	9%
1875	16,019,917	6,829,869	83,821,538	19%	8%
1876	16,113,200	6,665,762	86,028,263	19%	8%
1877	16,345,152	6,426,474	94,844,243	17%	7%
1878	16,891,776	6,645,984	87,871,463	19%	8%
1879	17,423,316	6,621,397	87,633,281	20%	8%
1880	17,817,636	6,727,321	94,793,781	19%	7%
1881	18,281,799	6,859,771	99,857,118	18%	7%
1882	18,514,701	6,971,356	119,243,012	16%	6%
1883	15,684,020	6,989,899	113,699,209	14%	6%
1884	19,092,183	6,968,400	112,829,860	17%	6%

Note: Figures in mln Lire (current prices). Disaggregated data are not available for the early years.

ruota, needy parents would have left their babies in the streets or public squares. Economic considerations played an important role in favor of the abolition. Pro-abolition supporters stressed that the existence of the *ruota* represented a pull factor that dramatically increased the number of abandoned children, including legitimate children that without the anonymity granted by the *ruota* would never be accepted in foundling hospitals. Naturally, rising numbers increasingly put the institutions caring for foundlings under severe financial stress. This was particularly burdensome for provincial councils and municipalities because the expenses to maintain foundling institutions were put on their shoulder according to the Law of 20 March 1865.²⁰ References to the issue of excessive expenses were frequent in the minutes of provincial councils discussing the matter and reports on laws proposed to address the matter (Scotti, 1877; Nicotera, 1878).

In 1868, provincial councils spent 13.9 mln Lire (22% of total expenses) on charitable activities, of which maintaining foundlings was the main item, accounting for more than half of all charitable expenses. As Table 1 shows, in the subsequent years, the expenses on foundlings were first reduced and then stabilized while charity and total expenses increased. Replacing the anonymity of the *ruota* with a consignment office where the origin of the baby could be verified was seen as the first policy to implement, along with a thorough reform of the foundling institutions.

The report from a parliamentary commission that studied the problem of provincial finances in the early 1870s summarizes well the main economic considerations that concerned the provincial councils:

“Even more serious is the issue of foundlings. Between the provinces and municipalities, they cost twelve million a year. Their number does not show any sign of decreasing. Finding a way to reduce or at least not further increase this expense is necessary. We believe the first reform towards this goal should be the abolition of the foundling wheel”.

To reinforce their argument, the commission reported the examples of the provinces where the *ruote* had already been abolished, which showed positive results in many respects:

“The abolition of the foundling wheel reduced the number of abandoned children, firstly because it excludes legitimate children, who unnecessarily enter in large numbers through the wheel system into foundling homes; secondly, because it serves as a salutary preventive check that, by removing the hope of secrecy, it enhances the power of modesty and lessen the illusions stemming from the guilt.

¹⁸ Mortality among foundlings was much higher than the child population (Kok et al., 1997; Rollet, 1997; Da Molin, 2002). This is true in the international context and has been studied in London Foundling Home (Levene, 2005), in Madrid's *La Inclusa* (Revuelta-Eugercios, 2013), in Normandy (Bardet et al., 1997). In the Italian case, this pattern was investigated by the work of Kertzer and his co-authors, who focused their attention on the area of Bologna in the nineteenth century (Kertzer and White, 1994-12; Kertzer, 1999-03-01; Kertzer et al., 1999-01) and of Corsini (1987), first, and Viazzo and his co-authors Viazzo et al. (1997), then, who instead focused on Tuscan and Florentine foundlings. Infant mortality rates varied substantially across Italian foundling homes (Kertzer et al., 1999-01), ranging from 300 to 600 in the nineteenth century, which was much higher than infant mortality in the general population, which, according to Del Panta (1997) was 270 in the aftermath of the country's Unification. In the foundling homes of Cosenza (Tocci, 1878; Caravetta, 2009) and Naples (De Crescenzo, 1873; Da Molin, 1991), infants died at a horrific rate. In particular, in Naples, infants were taken care of inside the overcrowded foundling home due to the lack of funding to pay for external wetnurses, thus favoring the spread of diseases and further increasing the mortality rate.

¹⁹ Available statistics on the 1880s, however, do not support this hypothesis, as recorded infanticides and abortions actually declined over this period (MAIC, 1888).

²⁰ It is important to note that the majority of the population did not elect provincial councils, but before 1883 only 2% of the male population was entitled to vote, as voting rights were based on how much taxes were paid. Therefore, even if the abolition was pursued for electoral reasons, there was no incentive to do it in the interest of the vast majority of the population. In this respect, the abolition process can be considered orthogonal to the preferences of the 99% population (both males and females).

Table 2
Variables, time span, and sources.

Variables	Time span	Source
Births (males, females, total)	1863–1882	Movimento dello Stato Civile
Illegitimate births (males, females, total)	1863–1882	Movimento dello Stato Civile
Foundlings (males, females, total)	1863–1882	Movimento dello Stato Civile
Stillbirths (males, females, total)	1863–1882	Movimento dello Stato Civile
Deaths (total)	1863–1882	Movimento dello Stato Civile
Infant mortality (males, females, total)	1863–1882	Movimento dello Stato Civile
Marriages	1863–1882	Movimento dello Stato Civile
Total Expenses (Province)	1863–1882	Bilanci comunali e provinciali
Total Revenues (Province)	1863–1882	Bilanci comunali e provinciali
Total Expenses (Municipalities)	1863–1882	Bilanci comunali e provinciali
Total Revenues (Municipalities)	1863–1882	Bilanci comunali e provinciali
Illiteracy of newlyweds	1866–1882	Movimento dello Stato Civile
Number of <i>ruote</i>	1863	Annali di Statistica (1884)
Electoral turnout	1866	Annuario delle province Italiane (1873)
Clergy (males)	1871	Italian population census
Clergy (females)	1871	Italian population census
Wet-nurses	1871	Italian population census
Midwives	1871	Italian population census
GDP	1871	Chiaiese (2024)
Population	1871	Chiaiese (2024)
Female literacy	1871	Ciccarelli and Weisdorf (2019)
Industry value added	1871	Canning and Schultz (2012)

And a reduction in number means a reduction in expenses. Naturally, the decrease in number, which in some places was immediate, significant, and absolute, in others progresses gradually. It is evident that the figure is lower than it would have been if the abolition of the wheel had not intervened to interrupt the course of previous increases.

The decrease in the number of abandoned children not only means an absolute or relative reduction in expenses but also affects their mortality rate. While it is painfully astonishing that, under the wheel system, mortality could exceed even 90 percent, after the abolition of the wheel, it decreased so significantly that in just a few years, it dropped from 70 to 30 percent in Turin and from 19 to 11 percent in Milan. [...] This is not surprising, as with fewer children, it is easier to select good wetnurses and provide more compassionate care for their upbringing”.

Several important points emerge from the report. Firstly, the increasing number of abandoned children was a financial problem for the newly-created provinces. Secondly, contemporaries acknowledged that the anonymity of the wheel encouraged the abandonment of children, representing a strong pull factor. Thirdly, they saw the abolition of the *ruota* and the end of the anonymity system as a solution to this problem. On the one hand, it made the abandonment of legitimate children impossible. On the other hand, they thought that the existence of anonymity affected people’s behavioral incentives by removing the natural brakes that “modesty and guilt” should provide. Finally, they observed that infant mortality among foundlings decreased where the *ruota* had been abolished and ascribed it to improved care for foundlings.

3. Sources

The main data contribution of this paper is a new annual panel dataset of Italian provinces in 1863–1882. Our data is at the provincial level, as foundling systems were devised at this level.²¹ The main source employed was the Civil Register of the Italian population (*Movimento dello Stato Civile*), which was published by the Directorate of Statistics of the Ministry of agriculture, industry, and trade (*Ministero dell’industria, dell’agricoltura e del commercio*, henceforth MAIC). Data were mostly hand-collected by the authors, but OCR techniques were implemented when feasible. Table 2 presents both the new variables digitized at the provincial level and the whole set of controls. For each variable, it reports the time coverage we consider and the source. Appendix A provides a detailed description of all the variables in the dataset.

The year 1863 marks the inaugural publication of Civil Register data by Italian authorities, thereby establishing the first year for which data can be included in our dataset. We decided to cut our analysis by 1882 because, in the 1880s, several inquiries by the Italian State were likely to have affected the institutional setting of managing child abandonment. Firstly, 1883 was the year the Italian State carried out the first comprehensive inquiry on the conditions of foundling hospitals. Even though a comprehensive reform was not enacted immediately, it is possible that the State’s formal inquiry affected local institutions’ practices, especially regarding high mortality among foundlings — possibly fearing repercussions and sanctions. Furthermore, it was followed in 1885 by the general inquiry on health and hygienic conditions of Italian municipalities, which dedicated a full section of the report on

²¹ The available data sources do not provide sufficient information at the municipality level for this period. Specifically, all the outcomes and some control variables are only available at the provincial level.

the issue of child abandonment. A final rationale for the cessation of our analysis in 1882 was the alteration of the methodology employed for collecting data from the Civil Register in 1883.²²

Since the first years following the Unification, the Civil Register has been used to record each municipality's births, deaths, and marriages. In 1863, the unitary government issued a circular to all prefects with instructions on how to register these events. The circular was addressed to parish priests, clergymen, mayors, and members of municipal and provincial councils. Local authorities were required to send monthly municipal summaries to the General Statistics Division (ISTAT, 2011).

Special attention was paid to the registration of foundlings, with the aim of overcoming the exclusion of foundlings from statistics by some pre-unitary administrations. The summary volume of the Civil Register for the years 1862–1878 outlines certain problems related to the registration of foundlings. For the period under consideration, births are classified into three categories: legitimate, illegitimate, and “exposed” (in Italian, *esposti*). The term “exposed” refers to abandoned children whose status is unknown. Italian statisticians were aware of the potential concerns related to this category, who may have been registered twice, before and after being abandoned. However, this concern may affect both the localities where the wheel is still open and those where it has closed. Starting in 1884, the method of recording foundlings changed, consolidating the last two categories. This change in labels provides another rationale for the cessation of our analysis in 1883.

We focus our analysis on four main outcome variables. These include the number of foundlings, births and infants who died within the first year of life.²³ All variables are disaggregated by gender. The fourth outcome variable we consider is the gender gap in abandonment, defined as the ratio of abandoned girls to boys by province and year. In Appendix F, we use an alternative outcome variable, that is yearly weddings. We complement this data with additional sources to add important control variables to our dataset. We collect total expenditures and revenues for all provinces and the total expenditures and revenues of all municipalities (aggregated at the provincial level) for the period 1863–1882.²⁴ We collect these variables from official publications, *Statistica del Regno d'Italia*, (1868, 1874) and MAIC (1879b, 1884b, 1886b). We then calculate the expenses and revenues per capita for each year.²⁵ We also collect data on weddings (1863–1882) and illiteracy of newlyweds from the Civil Register (1866–1882), number of stillbirths disaggregated by gender (1863–1882), and deaths. Plus, we collect data on clergy, wetnurses, and midwives from the 1871 census (MAIC, 1876).

Table 3 presents descriptive statistics for the main variables in the dataset, reporting the mean, median, standard deviation, minimum, and maximum for each outcome variable (expressed both in log and in absolute terms) and for the set of controls. The upper panel reports the results for the provinces that, by the end of the period, had received the treatment. The lower panel reports the results for those who did not. The discrepancies between the two samples are readily apparent. All the outcome variables exhibit higher values in the treated provinces. These differences are likely to be driven by pre-existing differences between treated and never-treated provinces. Nevertheless, the outcome variables expressed in the logarithmic form are observed to be lower in the treated sample. This may indicate that the relative decrease is more pronounced in the initial sample than in the subsequent one. The gender ratio in abandonment, defined as the ratio of males to females in instances of abandonment over the entire period, is higher in the treated provinces. A gender ratio in the abandonment of 106.25 is aligned with that of 106.3 observed in the general population during the Liberal period (Beltrán Tapia and Cappelli, 2024). The lower ratio of abandonment of 104.31 in the untreated provinces indicates a sex-selectivity bias in abandonment, where male abandonment outnumbered female abandonment. The two samples also differ according to the control variables. The treated provinces were generally wealthier, with fewer clergymen per capita, and exhibited higher female literacy and less illiteracy among newlyweds. Municipal expenditure was, on average, higher, whereas provincial expenditure was lower, suggesting differences in the share of responsibilities between municipal and provincial authorities. It is, therefore, necessary to add controls to the model before comparing the effect of the treatment on the outcome variables. Given the aforementioned differences between the two samples, one of our main specifications uses only not-yet treated provinces as control.

Concerning the treatment, we use official reports on infant abandonment produced by the MAIC to obtain information on the number of *ruote* by province and year and the year of the abolition of the *ruota* by province. In this endeavor, the main data source we use is a report published by MAIC, featuring a list of all the municipalities and provinces where the *ruota* had been abolished or was still existing on the 31st December 1877 (MAIC, 1878). Subsequently, we integrate the information on the closures with subsequent reports, most notably the State's first inquiry on the conditions of the foundling hospitals (MAIC, 1884). These allow us to construct our treatment variable, a dummy that, by each province and year, takes on value zero if the *ruota* was still working and value one if it had been abolished. However, as Fig. 1 shows, not all provinces had only one *ruota*, with Southern provinces featuring almost a *ruota* by municipality. Nonetheless, in many provinces, especially in the North and in the Center-North, each province featured a main one. An example is the case of Florence, where despite the *ruote* scattered around the area, child abandonment was centered around the city's foundling home, the *Ospedale degli Innocenti*. Often, all the *ruote* in the same province were closed on the same date. When the closure dates differed, the treatment changed to one when the provincial capital abolished the *ruota*. Appendix B reports the abolition dates for all provinces and describes how we selected a single date for those featuring more than one *ruota*.

²² Until 1882, provinces collected and processed the municipality data, aggregated them and then sent the resulting tables to the Directorate General of Statistics at MAIC. From 1883 onward, the data were processed and aggregated by the Directorate General of Statistics itself (MAIC, 1884).

²³ The choice of focusing on this particular threshold (1 year) to define infant mortality is not arbitrary. We use the same threshold category adopted by the general inquiry on the health and hygienic conditions of the Italian population (MAIC, 1886).

²⁴ The source provides data on municipal expenditures and revenues aggregated at the provincial level, representing the sum of all municipalities within each province.

²⁵ Annual population is calculated using linear interpolation between census years.

Table 3
Summary statistics — Treated and not treated.

	Mean	SD	Median	Min	Max	N
Treated provinces						
Foundlings	994.35	945.97	671.50	76.00	7139.00	806.00
Foundlings (log)	6.56	0.83	6.51	4.33	8.87	806.00
Births	14828.53	8914.23	12953.00	1486.00	45518.00	831.00
Births (log)	9.43	0.61	9.47	7.30	10.73	831.00
Infant mortality	3247.32	1918.82	2813.50	414.00	11101.00	774.00
Infant mortality (log)	7.91	0.61	7.94	6.03	9.31	774.00
Gender abandoned ratio	106.25	14.11	105.10	73.68	245.48	806.00
Marriages (log)	7.82	0.66	7.89	5.76	9.17	831.00
Deaths (log)	8.89	0.61	8.89	7.20	10.30	772.00
GDP	172.22	115.69	129.69	51.42	511.30	840.00
GDP (log)	4.95	0.62	4.87	3.94	6.24	840.00
Population	4.1e+05	2.4e+05	3.6e+05	1.1e+05	1.0e+06	840.00
Population (log)	12.75	0.59	12.79	11.58	13.83	840.00
Clergymen per capita	0.41	0.16	0.38	0.17	1.00	840.00
Female literacy	0.33	0.17	0.31	0.04	0.76	840.00
Municipal expenditure per capita	15.90	9.77	13.17	2.92	89.71	758.00
Provincial expenditure per capita	2.81	1.34	2.67	0.09	13.24	742.00
Illiteracy of newlyweds	0.58	0.18	0.60	0.14	0.92	700.00
Midwives and wetnurses per capita	0.00	0.00	0.00	0.00	0.00	840.00
Never-treated provinces						
Foundlings	889.89	424.08	884.50	122.00	3049.00	536.00
Foundlings (log)	6.66	0.54	6.79	4.80	8.02	536.00
Births	13880.66	5615.56	12732.00	5426.00	31134.00	540.00
Births (log)	9.46	0.38	9.45	8.60	10.35	540.00
Infant mortality	2987.95	1236.70	2753.00	934.00	7814.00	511.00
Infant mortality (log)	7.92	0.41	7.92	6.84	8.96	511.00
Gender abandoned ratio	104.31	10.31	103.63	67.45	195.24	536.00
Marriages (log)	7.84	0.48	7.83	6.27	9.05	540.00
Deaths (log)	8.91	0.42	8.86	7.89	10.15	510.00
GDP	125.73	45.83	114.08	61.90	224.54	540.00
GDP (log)	4.77	0.34	4.74	4.13	5.41	540.00
Population	3.6e+05	1.4e+05	3.2e+05	1.6e+05	7.0e+05	540.00
Population (log)	12.72	0.36	12.68	12.00	13.46	540.00
Clergymen per capita	0.48	0.09	0.48	0.27	0.69	540.00
Female literacy	0.13	0.07	0.10	0.05	0.41	540.00
Municipal expenditure per capita	10.96	4.46	10.56	3.02	24.27	509.00
Provincial expenditure per capita	3.09	1.71	2.72	0.22	14.85	506.00
Illiteracy of newlyweds	0.78	0.09	0.81	0.49	0.94	458.00
Midwives and wetnurses per capita	0.00	0.00	0.00	0.00	0.00	540.00

We have constructed an alternative treatment that excludes partially treated provinces — those where some, but not *all*, the *ruote* were closed on the same date. In particular, we consider as partially treated only those provinces which still had some *ruote* open by 1882. The alternative treatment only changes to one when all the province's *ruote* were abolished. In doing so, we exclude all the partially treated provinces from our sample. The last column (Partially) in [Appendix B](#) indicates whether each province is partially treated. Using this alternative treatment permits to rule out the possibility of same province municipal-level spillovers, driven by a person traveling to a neighboring municipality within the same province to abandon an unwanted baby.²⁶

4. Methodology

We study the effect of the abolition of the *ruota* (our treatment) that happened at different times (with a staggered timing) across Italian provinces. It should now be clear that our treatment was heterogeneous across provinces, not just in its staggered timing but also in its dissemination from Northern to Southern provinces. According to the econometrics and applied literature, standard two-way fixed effects (TWFE) diff-in-diff estimators, even when allowing for dynamic treatment effects, are likely biased and produce significant estimates with the wrong sign ([Baker et al., 2022](#)). Heterogeneity in treatment effects is likely to characterize most economic settings, thus producing biases in research settings with staggered timing tackled with standard diff-in-diff estimates.

We adopt a semi-parametric diff-in-diff estimation method implemented by [Callaway and Sant'Anna \(2021\)](#) to address these challenges. Heterogeneous diff-in-diff allows for the treatment effect to vary over time and across cohorts. It first estimates the

²⁶ Concerning border municipalities, we cannot rule out the problem of traveling to a municipality in a neighboring province. However, one should consider the high costs of traveling at the time, which would make it unfeasible for poor families, who are more affected by the abolition. Therefore, we do not expect this phenomenon to be substantial in magnitude.

individual cohort time-specific treatment effects, allowing for treatment effect heterogeneity, and then aggregates them to produce a measure of the overall treatment effects, the aggregated Average Treatment Effects on the Treated (henceforth, ATT). In our staggered difference-in-differences framework, the ATT measures the average causal effect of the treatment on those units that received the treatment. By comparing the treated units to appropriate control units before and after the staggered implementation of the treatment, we isolate the effect of the treatment from other temporal changes. It is a single number that is easy to interpret as it is comparable to the regression coefficient of a standard diff-in-diff framework. Unlike that, the single ATT is a weighted average across cohorts considering each treatment cohort's sample share. Among the set of staggered diff-in-diff estimators, we use the augmented inverse probability weighting (AIPW) estimator. The AIPW is used to adjust for confounding variables because of its double robustness properties. In fact, it combines the properties of regression-based and inverse probability estimators. The first step consists of fitting a propensity score model; the second step estimates the outcome under treatment and control conditions. Unlike other staggered diff-in-diff estimators, such as the inverse-probability weighting (IPW), the regression-adjusted (RA) (Callaway and Sant'Anna, 2021), and the extended TWFE (Wooldridge, 2021), it permits to correctly specify both the treatment and the outcome models by adding a series of controls. The double robustness property allows us to misspecify one of the two models while still getting consistent estimates. Moreover, a distinctive feature of this set of estimators is that it allows the use of never-treated units and not-yet-treated ones as well as a control. In the specification presented in the main body of the paper, we use both samples as controls. In one of the specifications presented in the results table, we use only the not-yet-treated provinces as a control, excluding the never-treated units from the sample.

All staggered diff-in-diff estimators include unit (in this case provinces) and time fixed effects. The first estimation step involves fitting a propensity score matching model to account for the probability of treatment assignment based on observed time-invariant characteristics. This set of controls used for the propensity score matching is defined as treatment model. Including it permits to circumvent the potential risk that the effect of the *ruota* was due to some pre-existing trait. The controls included are a dummy variable for Southern Italy, considering the higher probability of *ruota* closures among Northern provinces, aimed at preventing the effect from solely reflecting a North–South difference. A proxy for religiosity, measured as the number of clergymen per capita at the province level obtained from the 1871 population census, is included to capture the Catholic Church's varying influence on *ruota* closures (MAIC 1876). Additionally, we control for female literacy as it may indicate a progressive attitude that could influence the timing of the treatment (Ciccarelli and Weisdorf, 2019), for the number of wetnurses and midwives per capita to capture the *ex ante* heterogeneity in the abandonment system that might have had a continued impact on the abolition. Finally, we incorporate a control for GDP per capita in 1871. The second estimation step involves a regression-adjusted estimation. As controls, we include variables that could influence the outcome variable. This set of controls, defined as the outcome model, is considered in $t-1$, that is, the year before the adoption of the treatment. The control variables are the total number of births, deaths older than one year, marriages, municipality and province expenses, and revenues per capita, as we argue, which might impact the investment in children.

For each outcome variable, we show a table reporting the ATT from six different specifications. The first specification (Baseline) does not include any control variable. The second specification (TWFE) reports the results of a standard TWFE. The third specification (Treatment Model) performs the propensity score matching (it includes what has been defined as the treatment model), but does not include controls on the outcome. The fourth specification (No Partially) replicates the analysis with the full set of controls on a restricted sample, which discards the partially treated provinces. The fifth specification (Not-yet only) uses the full control set but considers only the not-yet treated variable to be the control group. In these two specifications, we do not include the dummy for the South because we lose between 29 and 49% of the observations, many of which are in the South, and the model does not achieve convergence. The last specification (Full model) includes all the controls (treatment and outcome model).

We then show the dynamic event study plots obtained from our preferred specification (the last one) containing pre-treatment estimates that can be used to pre-test the parallel trend assumption and treatment effect estimates in post-treatment periods. To assess the robustness of our findings, in Appendix E, we perform placebo tests by randomizing the timing of treatment adoption across the sample. Specifically, we assign the treatment to different units (also to the never-treated units) at random time points, creating a scenario where no actual treatment effect should be observed if the estimated effects are genuinely attributable to the treatment rather than spurious correlations or time trends. By comparing the results from these placebo tests with our main findings, we can evaluate whether the observed treatment effects are statistically significant and not driven by random chance or other confounding factors.

The parallel trend assumption (henceforth PTA) in a staggered context must be discussed. In general, the PTA requires that the difference between the treated and the control groups is constant over time in the absence of the treatment. In a staggered context, the assumption of parallel trends is substantially more plausible if it holds after conditioning on observed pre-treatment covariates. This is what we do in the first step of the estimation.

5. Results

5.1. Abandonment, births, and infant mortality

This paper aims to examine the evolution of the treatment effect in the post-treatment period. To do so, we estimate the following model using Callaway and Sant'Anna (2021)'s estimator:

$$Y_{it} = \alpha_i + \lambda_t + \tau_{it}D_{it} + \beta X_{it} + \varepsilon_{it}, \quad (1)$$

Where:

Table 4
Effect of the abolition of the *ruota* on infant abandonment.

Model description	Twfe	Baseline	Treatment	No Partially	Not-yet only	Full model
ATT	−0.547	−0.767	−0.669	−0.635	−0.514	−0.780
S.E.	0.132	0.210	0.134	0.148	0.106	0.149
p-val	0.000	0.000	0.000	0.000	0.000	0.000
obs	1185	1331	1331	843	607	1176
Outcome Model Controls:						
Log births	yes	no	no	yes	yes	yes
Log deaths (>1 y.o)	yes	no	no	yes	yes	yes
Municipality expenses per capita	yes	no	no	yes	yes	yes
Province expenses per capita	yes	no	no	yes	yes	yes
Treatment Model Controls:						
Female literacy (1871%)	no	no	yes	yes	yes	yes
Wetnurses and midwives per capita	no	no	yes	yes	yes	yes
Log GDP (1871)	no	no	yes	yes	yes	yes
Log Population (1871)	no	no	yes	yes	yes	yes
Male Clergy per capita	no	no	yes	yes	yes	yes
Dummy South	no	no	yes	no	no	yes

Notes: The models present various specifications showing the effect of the abolition of the *Ruota* on child abandonment rates. Twfe represents the Two-Way Fixed Effects model. The models control for different combinations of covariates, as specified in the Outcome Model and Treatment Model sections.

Interpretation: The full model estimate suggests a treatment effect of approximately −0.780 on the log of the number of abandoned children. This implies a reduction of about 54.2% in the abandonment rate ($e^{-0.780} \approx 0.458$). The controls included in the different models ensure robustness and check for various potential confounders.

- Y_{it} is the outcome variable for province i at time t
- α_i are province fixed effects
- λ_t are time fixed effects
- D_{it} is the treatment indicator (1 if province i is treated at time t , 0 otherwise)
- τ_{it} is the heterogeneous treatment effect, which can vary by province and time
- X_{it} is a vector of control variables
- β is a vector of coefficients for the control variables
- ε_{it} is the error term.

As shown in Table 4, we start by examining the effect of the abolition of the *ruota* on the number of abandoned infants. Infant abandonment is measured as the log of the children reported as *esposti* (literally exposed, foundlings). The ATT is negative and statistically significant at the 1% level across the six specifications, thus confirming that the abolition of the *ruota* led to a drop in the number of foundlings. The ATT ranges from −0.547 to −0.780 across the specifications. In the full model, reported in the last column, the ATT is −0.780. This means the number of abandoned children is roughly 0.458 times what it would be without the treatment, or a reduction to about 45.1% of the original number. The overall effect is a reduction of 54.9% throughout the period. The result holds when we discard partially-treated units, and we find an overall effect of 47%, and when we consider only not-yet treated units as a control. In the latter case, the effect of the abolition of the *ruota* on abandonment is smaller, 40.2% but still statistically significant. Fig. 2 illustrates the dynamic impact of the *ruota*'s abolition on the count of abandoned children in the full model. The graph validates the absence of a pre-trend and exhibits a significant negative effect on the number of abandoned children in the post-treatment period. Considering our treatment's imposition of a ban on legitimate parents from abandoning their children, we witness an immediate decline in abandonment at time zero. The effect intensifies with prolonged exposure to the treatment, indicating a stronger impact the longer the duration of exposure. However, the effect loses statistical power over time. The right-hand side panel in Fig. 12 in Appendix E shows the event study with randomized treatment. As the p -value on the ATT shows, the effect of the abolition is not statistically significantly different from 0.

Appendix C shows that our finding is robust when additional control variables are incorporated into the outcome (Table 10) and treatment models (Table 11). The second column of Table 10 considers the combined effect of provincial and municipal expenditures, thus avoiding the potential bias that may arise from the different sources of funding (municipality or province). The third column incorporates a measure of the illiteracy of the spouses, based on their ability to sign the marriage certificate. The fourth column considers expenditures in their absolute value and incorporates a control for population. The fifth column excludes expenditure and includes municipal and provincial revenues per capita. The sixth specification considers stillbirths, reflecting the hypothesis that they may conceal abortions and infanticides. The seventh column includes weddings among the control. The eighth does not control for deaths, whereas the final specification controls for all deaths, regardless of age. The ATT varies only slightly and remains significant at the 1% level. Concerning the robustness checks on the treatment model, column two of Table 11 considers female clergy per capita rather than male component. The third column employs an alternative treatment variable, which identifies the treated group as provinces where the *ruota* was abolished, and legitimate abandoned infants outweighed illegitimate ones. Consequently, the control group encompasses provinces where the *ruota* remained operational and provinces where the *ruota* had ceased to operate,

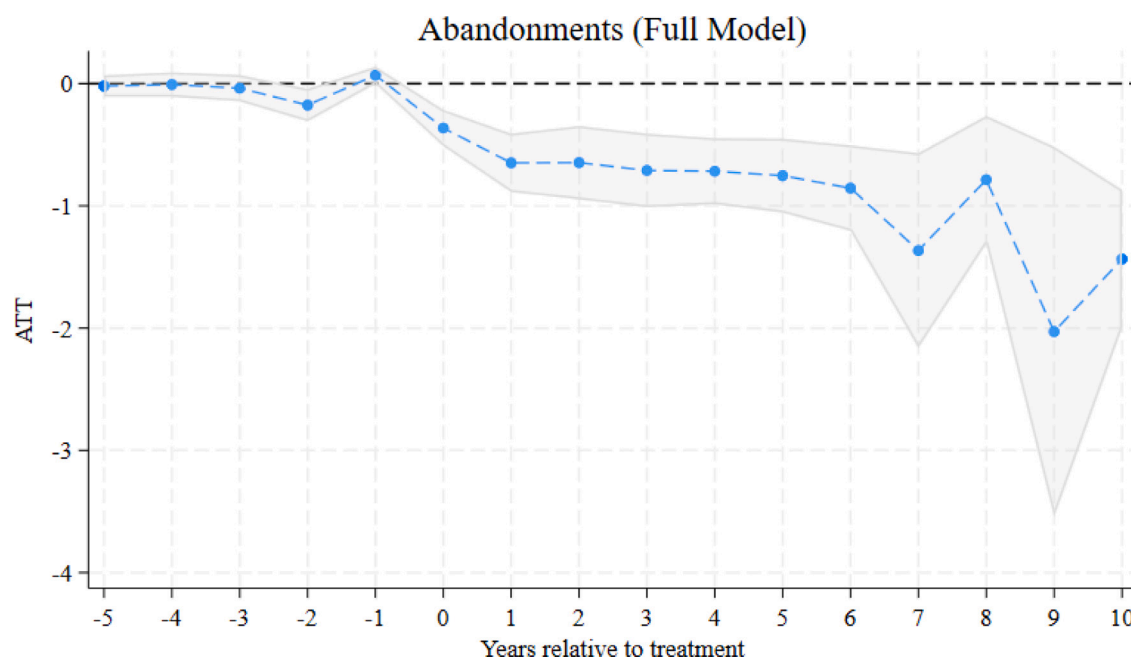


Fig. 2. Dynamic event study plot: the effect on infant abandonment.

but illegitimate abandonment was more common. The effect is even more pronounced in these provinces. The fourth column controls for the number of *ruote* by province at the beginning of the period to capture an additional dimension of pre-existing differences in the abandonment system by province. The fifth column considers the turnout at the 1866 elections, which can be measured as the share of eligible voters who actually cast their ballots, as a different involvement in the public and political lives of the province is likely to mirror divergent attitudes toward closure. The sixth column incorporates industrial value added, while the seventh excludes Southern provinces.

Table 5 shows the effect of the abolition of the *ruota* on the number of births, measured in log. The ATT is always negative and statistically significant in all specifications, confirming that the abolition had a negative impact on the number of new births. The values of the ATT in all specifications range from -0.025 to -0.050 . The full model (last column) features an ATT of -0.042 , thus revealing the effect on the number of births was an overall decrease by 4.1% throughout the entire period with respect to what it would have been without the treatment. The ATT is significant and only slightly smaller in the sample which discards the partially treated provinces. In this case we find that the closure of the *ruota* leads to a reduction in births by 3.8%. The ATT is observed to reach its lowest value when only units that have not yet undergone treatment are employed as a control, indicating that the abolition of the *ruota* resulted in a reduction in births by 2.65%. This outcome is not unexpected, given that the treatment and control groups in this instance are considerably more homogeneous, both in terms of geography and institutions. Moreover, we expect that spillover effects among provinces with greater similarities may serve to attenuate the observed effect. Fig. 3 confirms that the pre-trend in the full model is not statistically different from 0. Concerning the post-trend period, the effect starts to be statistically significant at time 2, that is, two years after the adoption of the treatment. The effect suggests that, after the abolition, households have to come up with alternative solutions, and thus might change their reproductive behaviors. The time lag is not surprising considering that pregnancy lasts forty weeks, thus the impact cannot be immediate as in the case of abandonment. The effect becomes stronger the longer the exposure to the treatment, but it loses statistical significance in the last year. Fig. 13 in Appendix E shows the results when randomizing the treatment. The overall ATT is not statistically significant, nor is the trend plotted in the dynamic event graph. Appendix C shows that the result is robust when adding the aforementioned additional controls on the outcome (Table 12) and on the treatment model (Table 13).

Subsequently, we estimate the effect of closing the *ruota* on legitimate and illegitimate births separately. Our findings indicate that the decline was primarily driven by a significant reduction in marital fertility.

Over the 20-year period, there was a 3.6% decline in legitimate births compared to what would have been expected in the absence of the *ruota*'s abolition. The dynamic event study illustrated in Fig. 4 exhibits a pattern that is strikingly analogous to that observed in the incidence of births. This is to be expected, given that the mean ratio of legitimate births to total births in our sample is 0.93. We then proceed to examine the impact of the *ruota* closure on non-marital fertility. Our findings indicate a statistically significant increase in the number of illegitimate births over time, resulting in a 32% rise in the total number of recorded illegitimate births.

It is not our contention that the closure of the *ruota* resulted in an actual increase in non-marital fertility. Rather, we argue that this increase is primarily attributable to the manner in which abandonment was registered. As previously stated, the Civil Register

Table 5
Effect of the abolition of the *ruota* on births.

Model description	Twfe	Baseline	Treatment	No Partially	Not-yet only	Full model
ATT	−0.050	−0.027	−0.032	−0.039	−0.025	−0.042
S.E.	0.010	0.009	0.009	0.013	0.011	0.010
p-val	0.000	0.003	0.000	0.002	0.023	0.000
obs	1185	1360	1360	843	607	1176
Outcome Model Controls:						
Log deaths (>1 y.o.)	yes	no	no	yes	yes	yes
Municipality expenses per capita	yes	no	no	yes	yes	yes
Province expenses per capita	yes	no	no	yes	yes	yes
Treatment Model Controls:						
Female literacy (1871%)	no	no	yes	yes	yes	yes
Wetnurses and midwives per capita	no	no	yes	yes	yes	yes
Log GDP (1871)	no	no	yes	yes	yes	yes
Log Population (1871)	no	no	yes	yes	yes	yes
Male Clergy per capita	no	no	yes	yes	yes	yes
Dummy South	no	no	yes	no	no	yes

Notes: The models present various specifications showing the effect of the abolition of the *Ruota* on birth rates. Twfe represents the Two-Way Fixed Effects model. The models control for different combinations of covariates, as specified in the Outcome Model and Treatment Model sections.

Interpretation: The full model estimate suggests a treatment effect of approximately −0.042 on the log of the number of new births. Exponentiating this coefficient gives $e^{-0.042} \approx 0.959$, indicating that the treatment causes the birth rate to decrease to about 95.9% of its original value, corresponding to a reduction of approximately 4.1%. The controls included in the different models ensure robustness and account for various potential confounders.

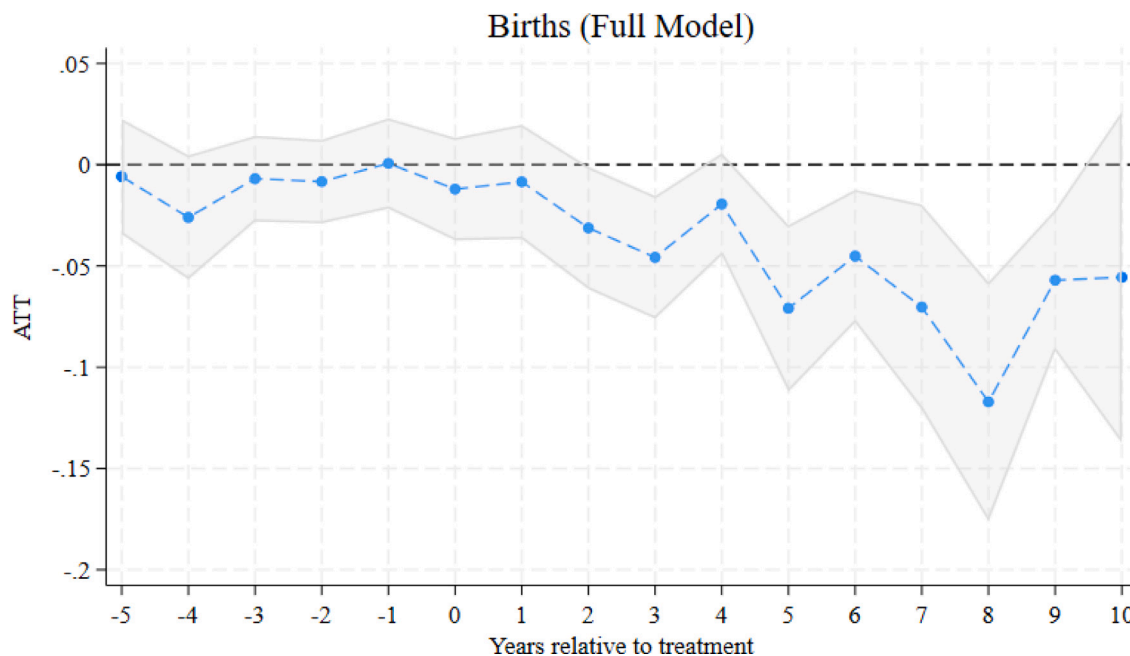


Fig. 3. Dynamic event study plot: the effect on births.

employs a tripartite classification system for births, categorizing them as legitimate, illegitimate, and “exposed”. As long as the *ruota* was in place and the identity of the parents unknown, the status of many infants who had been abandoned was unclear. Consequently, they were simply recorded as “exposed”. Following its closure, parents were required to provide their identity. This resulted in a genuine decline in legitimate abandonment, which was no longer condoned, coupled with an increase in the abandonment of illegitimate infants whose status was now known. Two elements support the registration hypothesis. The first, illustrated in Fig. 5, is that the impact on illegitimate births is positive and statistically significant at the same time as the treatment, without the anticipated lag due to the childbearing period. Secondly, the two categories, illegitimate and “exposed”, were employed as alternatives, a practice that can be traced back to the sources. Fig. 6 illustrates this phenomenon in the context of Milan. Following the abolition of the *ruota* in 1868 (represented by the black dotted vertical line), the number of children reported as “exposed” declined significantly. This trend is largely due to an actual decline in admissions, but also partly because newly abandoned infants

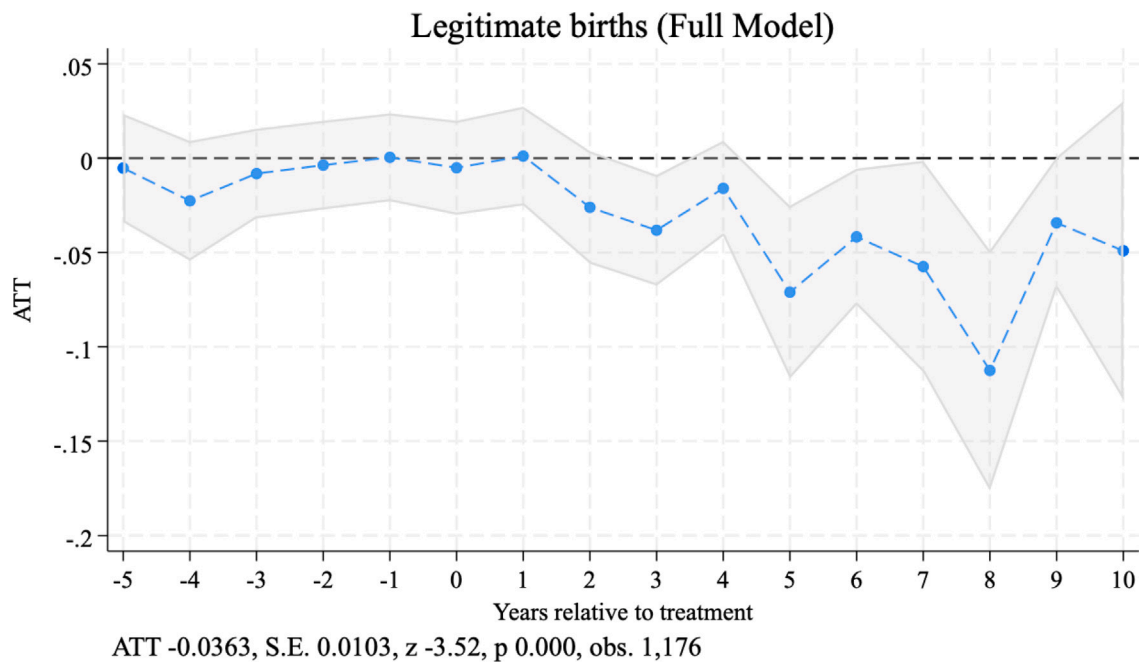


Fig. 4. Dynamic event study plot: the effect on legitimate births.

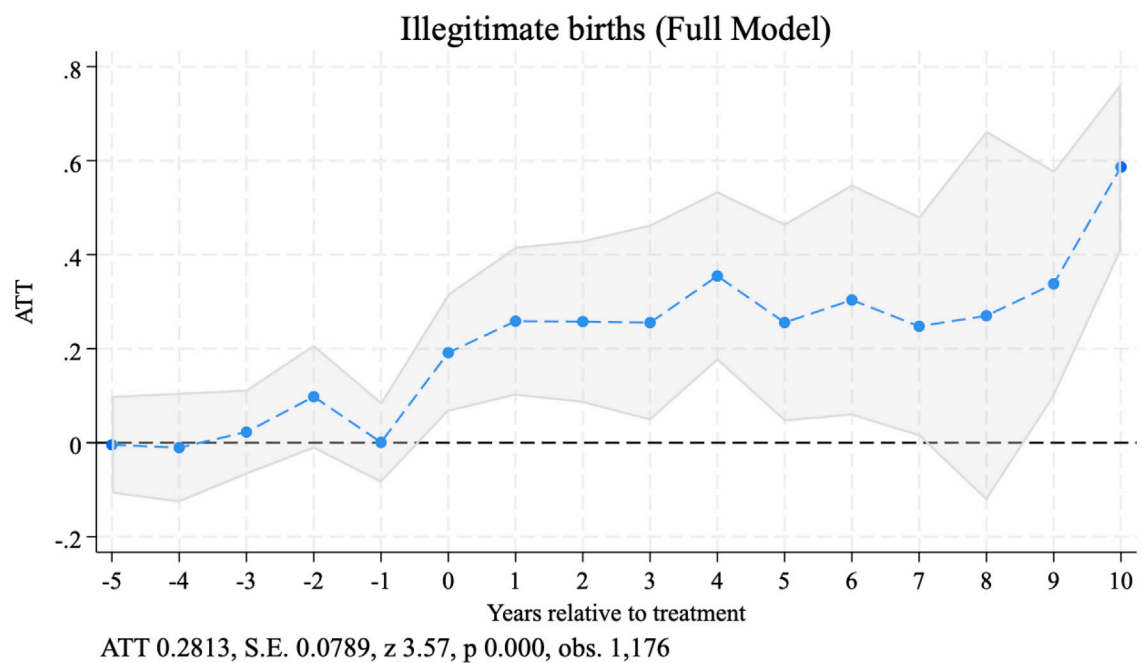


Fig. 5. Dynamic event study plot: the effect on illegitimate births.

are no longer recorded in this category and are instead classified as illegitimate. As the figure shows, illegitimate births (blue line) increase following the closure of the *ruota*. However, this rise falls far short of offsetting the decline in abandonment. In fact, when we consider the two categories together (green line), the impact of the *ruota*'s closure on the total number of non-legitimate births (combining illegitimate and exposed infants) diminishes considerably. The overlap between the green and red lines further confirms that, following the closure, *all* the abandoned children were indeed recorded as illegitimate.

We now extend our analysis beyond the descriptive case study of Milan to investigate the causal impact of the *ruota*'s abolition using the usual Difference-in-Differences (diff-in-diff) framework. As expected, Fig. 7 shows a negative and significant effect of on

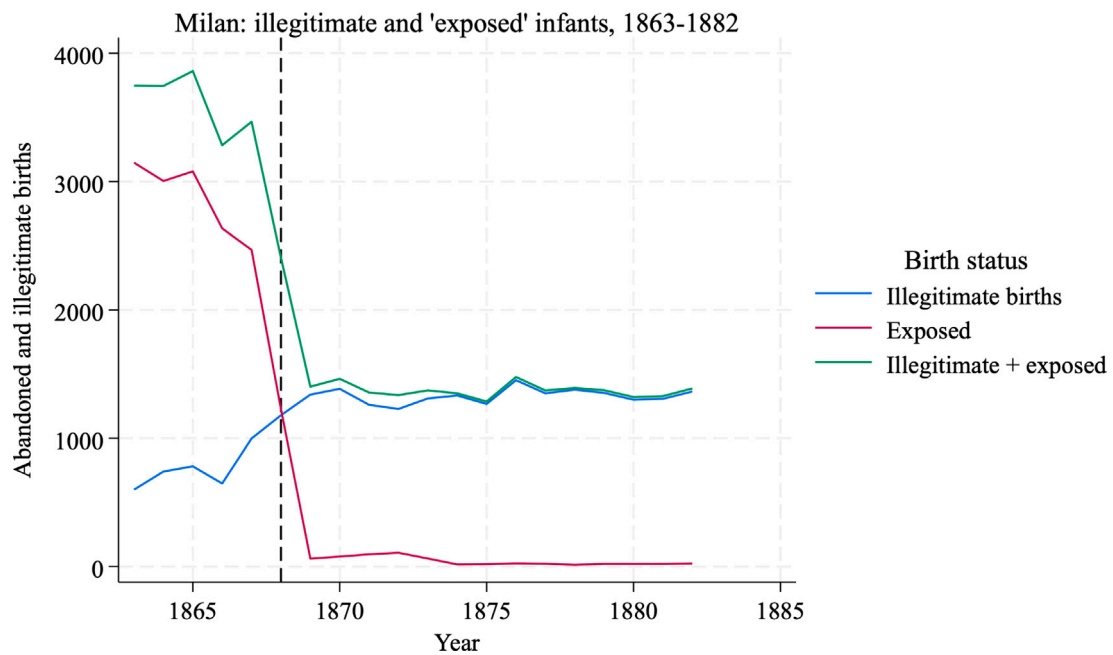


Fig. 6. Trend in illegitimate births and 'exposed' in Milan.

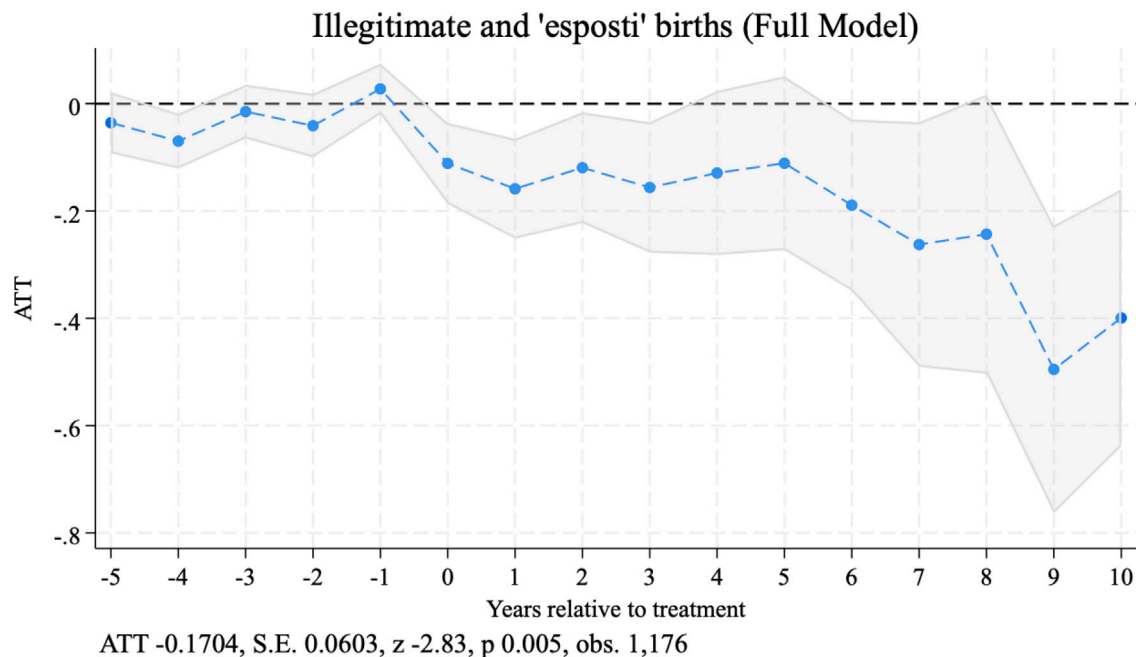


Fig. 7. Dynamic event study plot: the effect on illegitimate births and "exposed".

illegitimate and "exposed" considered together. Compared to the effect on the number of abandoned infant shown in Fig. 2, the effect here is smaller in magnitude and less significant. This is to be expected since it is the result of two contrasting effects: the positive and significant effect that the abolition of the *ruota* has on illegitimate births and the negative effect on "exposed" children. As a consequence, the strong negative effect on abandonment is "watered down".

Table 6 shows the effect of the abolition on infant mortality, measured as the log of infants who died in the first year of life. Also, in this case, the closure of the *ruota* is statistically significant and negative. In our preferred specification, the ATT is -0.106 , this means an overall reduction in infant mortality of 10.1% compared to the case in which the treatment had not been adopted.

Table 6
Effect of the abolition of the *ruota* on infant mortality.

Model description	Twfe	Baseline	Treatment model	No Partially	Not-yet only	Full model
ATT	−0.057	−0.074	−0.099	−0.051	−0.049	−0.106
S.E.	0.014	0.017	0.019	0.023	0.015	0.027
p-val	0.000	0.000	0.000	0.027	0.002	0.000
obs	1185	1274	1274	843	607	1176
Outcome Model Controls:						
Log births	yes	no	no	yes	yes	yes
Log deaths (>1 y.o.)	yes	no	no	yes	yes	yes
Municipality expenses per capita	yes	no	no	yes	yes	yes
Province expenses per capita	yes	no	no	yes	yes	yes
Treatment Model Controls:						
Female literacy (1871%)	no	no	yes	yes	yes	yes
Wetnurses and midwives per capita	no	no	yes	yes	yes	yes
Log GDP (1871)	no	no	yes	yes	yes	yes
Log Population (1871)	no	no	yes	yes	yes	yes
Male Clergy per capita	no	no	yes	yes	yes	yes
Dummy South	no	no	yes	no	no	yes

Notes: The models present various specifications showing the effect of the abolition of the *Ruota* on infant mortality rates. Twfe represents the Two-Way Fixed Effects model. The models control for different combinations of covariates, as specified in the Outcome Model and Treatment Model sections.

Interpretation: The full model estimate suggests a treatment effect of approximately −0.106 on the log of the infant mortality rate. Exponentiating this coefficient gives $e^{-0.106} \approx 0.899$, indicating that the treatment causes the infant mortality rate to decrease to about 89.9% of its original value, corresponding to a reduction of approximately 10.1%. The controls included in the different models ensure robustness and account for various potential confounders.

The closure of the *ruota* also has a statistically significant effect on the sample that excludes partially treated units (No Partially) and on the sample that considers only non-treated provinces as a control (Not-yet only), with respective effects of 4.9 and 4.8%. As with the data on births, the ATT is also lowest in the sample of provinces that have not yet been treated. The lower effect in the sample that excludes the partially treated provinces is explained by the fact that we are considering a province as treated only when it has abolished all its *ruote*. However, the effect is likely to start appearing even when only a few *ruota* have been closed. Fig. 8 shows that the pre-treatment is not statistically different from 0 and shows a negative effect after the adoption of the treatment. As in the case of births, the effect becomes statistically significant in the second year after the *ruota*'s closure, it stays constant until time 6 and starts increasing afterward. Once again, the lag in the effect is not surprising because we exclude that it might affect children who were already born at the time of adopting the treatment. The placebo specification and dynamic event study in Fig. 14 are not statistically significant. Appendix C shows the robustness of our findings across different specifications of the outcome (Table 14) and the treatment model (Table 15).²⁷

5.2. Gender discrimination in abandonment

We proceed to explore the potential connections between the closure of the *ruota* and the gender dimension of infant abandonment. We consider a specific dimension of gender disparity: the chances of being abandoned. Historical records indicate a bias in abandonment practices, where girls were more commonly abandoned than boys. This trend was notably prominent when married couples abandoned legitimate babies due to economic hardships. In such cases, households often favored keeping boys, seen as assets, over girls, who were perceived as liabilities, requiring dowries. On the contrary, illegitimate babies were abandoned, irrespective of their gender. Beltrán Tapia and Cappelli (2024) suggest that selective abandonment possibly contributed to the phenomenon of “missing girls” in Italy, linked to high mortality rates among foundlings. Consequently, we hypothesize that the *ruota*'s abolition and the subsequent decrease in legitimate abandonment might have influenced the gender bias in abandonment rates.

Table 7 confirms this hypothesis. While the controls on the treatment model are the same, we incorporate a slightly different set of controls on the outcome. We exclude controls on the municipal and provincial per capita expenses, as we argue they are unlikely to have affected the sex ratio. Alongside the number of births, we include the sex ratio in births, as it influences the sex ratio in abandonment, and the share of foundlings over total births, as we argue that the sex ratio in abandonment is influenced by how pervasive abandonment was. The ATT is positive across different specifications, confirming that the closure reduced the sex selectivity in abandonment. The impact of the closure of the *ruota* on sex selectivity in abandonment is not statistically significant when the never-treated units are excluded from the analysis. This is an unsurprising result, given the difficulty in detecting an

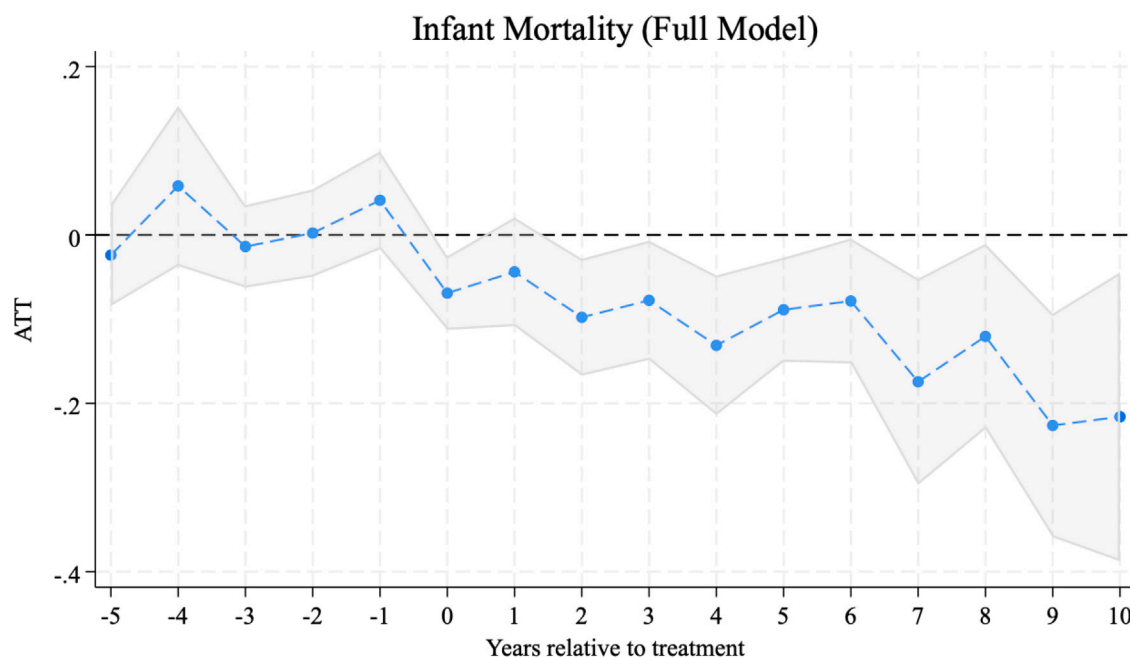
²⁷ When excluding the Southern provinces, the average treatment effect (ATT) loses statistical significance. However, plotting the event study reveals that the dynamic trend is consistent with the main specification, but confidence intervals are wider. This is not unexpected, given that we lost ca. 1/3 of our observations, and infant mortality is estimated using a smaller sample than foundlings and births.

Table 7Effect of the abolition of the *ruota* on gender discrimination in abandonment.

Model description	Twfe	Baseline	Treatment model	No partially	Not-yet only	Full model
ATT	1.292	7.127	13.953	12.918	2.813	13.964
S.E.	0.305	3.608	3.941	4.877	9.305	4.144
p-val	0.000	0.048	0.000	0.008	0.762	0.001
obs	1342	1331	1331	957	695	1331
Outcome Model Controls:						
Log births	yes	no	no	yes	yes	yes
Sex ratio	yes	no	no	yes	yes	yes
Foundlings-birth ratio	yes	no	no	yes	yes	yes
Treatment Model Controls:						
Female literacy (1871 %)	no	no	yes	yes	yes	yes
Wetnurses and midwives per capita	no	no	yes	yes	yes	yes
Log GDP (1871)	no	no	yes	yes	yes	yes
Log Population (1871)	no	no	yes	yes	yes	yes
Male Clergy per capita	no	no	yes	yes	yes	yes
Dummy South	no	no	yes	no	no	yes

Notes: The models present various specifications showing the effect of the abolition of the *Ruota* on gender discrimination in child abandonment. Twfe represents the Two-Way Fixed Effects model. The models control for different combinations of covariates, as specified in the Outcome Model and Treatment Model sections.

Interpretation: The full model estimate suggests an increase of approximately 13.964 percentage points in the ratio of male to female children abandoned. This suggests that the treatment reduced the relative frequency of female child abandonment compared to male abandonment, indicating a decrease in gender discrimination. The controls included in the different models ensure robustness and account for various potential confounders.

**Fig. 8.** Dynamic event study plot: the effect on infant mortality.

effect on the sex ratio and the profound influence of gender norms and institutions, as well as abandonment patterns, particularly the ratio of legitimate to total abandonment, on this outcome. Our argument is that the treated and not-yet-treated provinces were more similar in this regard prior to the adoption of the treatment, which explains why we do not find a statistically significant effect in this sub-sample. Fig. 9 shows the dynamic event graph for the full model. The pre-treatment is always not statistically different from 0, but at time -3 and -2 , where it takes on a different sign. The post-treatment trend consistently displays positive values and becomes statistically significant right after the treatment, losing statistical significance in some years, but always remaining positive. This observation implies that if abandonment indeed contributed to the phenomenon of “missing girls” in Italy, our quantitative analysis provides evidence that the *ruota*’s abolition alleviated this gender distortion. The fact that the result is not clearcut can conceal pre-existing differences in abandonment practices among provinces. Provinces themselves can be a unit too large to analyze

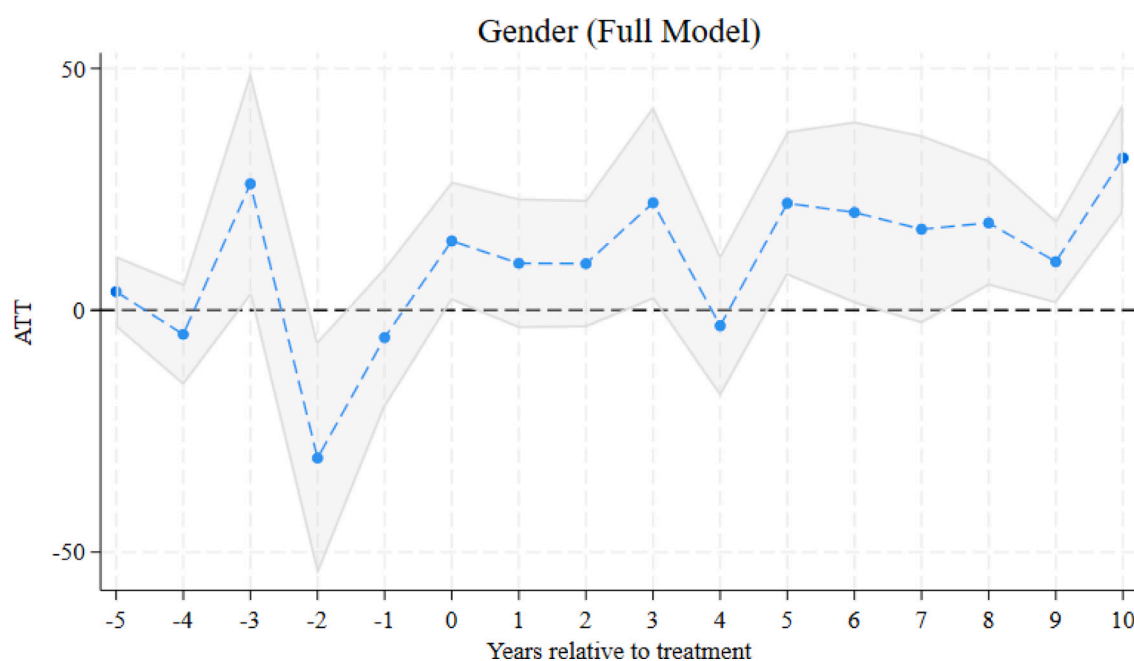


Fig. 9. Dynamic event study plot: the effect on gender discrimination in abandonment.

the phenomenon of missing girls. Future research is needed to explore the dynamics of sex ratio and sex ratio abandonment at a more granular geographical level.

The right-hand side panel in Fig. 15 in Appendix E confirms that the effect is not statistically different from 0 when the treatment is randomized. Appendix C shows that the result is robust across the standard set of additional control on the outcome (Table 16) and the treatment (Table 17) models.

Even if the abolition of the *ruota* had a positive effect on the gender bias in abandonment, leading to a relative reduction in female abandonment, we do not know what ultimately happened to the girls raised by their parents: Were they more deprived at home than their male counterparts? Was female infant mortality higher after the closure? We look at whether the abolition of the *ruota* affected the gender bias in infant mortality, and we find no effect (Fig. 11). The dynamic event study is presented in Appendix D. Moreover, even if foundlings represented a non-negligible proportion of the infant population, the abolition of the *ruota* did not have an impact on the sex ratio in the general population (Fig. 10 in Appendix D).

The absence of a statistically significant impact of the abolition of the *ruota* on gender equality may be attributed to the fact that the analysis has considered only one dimension of gender inequality, namely girls' premature death. Excess female mortality is an extreme example of gender inequality within the population, but not the only one (Schneider, 2016). Also, unequal distribution of food and care during infancy leads to inequalities in health outcomes. Despite the wealth of information in foundling home archives, anthropometric data is lacking for the nineteenth century, making it impossible to assess whether there was differential treatment by gender in terms of physical measures (Schneider, 2016). Existing studies have investigated accounts of (mal)nutrition and morbidity.²⁸

6. Discussion and conclusions

This paper has analyzed the effect of the abolition of the *ruota* in Italy two decades after the country's Unification. Using a staggered diff-in-diff approach, we find that the abolition of the *ruota* caused a reduction in abandonment, births, infant mortality, and sex selectivity in abandonment. We argue that the abolition was an institutional and social change that increased the cost of abandonment, possibly contributing to the fertility transition.

The first finding of the paper regards the decline in instances of child abandonment. The effect of the policy was to deter the abandonment of children born within wedlock. As expected, the impact of the policy on the number of abandoned infants is immediate and reflects the extent of compliance with the ban.

²⁸ For instance, Schneider (2023-04-03) examines the causal effect of diet (measured using anthropometric indicators) on the probability of contracting five infectious diseases common in childhood among children at the London Foundling Hospital for the period 1892–1919. The author finds no direct effect on the likelihood of contracting the disease but a statistically significant effect on the duration of the illness.

The second finding has to do with the decline in total births. As the high number of abandoned infants suggests, the *ruota* represented a form of preventive check commonly used in pre-transitional Italy. Following its closure, impoverished married households had to find solutions “in the bedroom” to control their fertility which, unlike abandonment, were likely to have an impact on the number of births (Cinnirella et al., 2017). The effect of the closure of the *ruota* on the diffusion of birth controls has already been inferred by historians of abandonment. According to Kertzer, Italians — particularly those residing in Northern cities — began recurring to birth controls after the *ruota*’s closure. First, we investigate whether a corresponding reduction in births could have driven the observed decline in abandonment. We rule out this possibility, as the two effects happened at different times, given that the impact on abandonment is immediate, whereas that on births occurs with a two-year lag. Secondly, in Appendix H we employ the staggered diff-in-diff approach, utilizing the proportion of exposed births as the dependent variable, rather than the absolute number of births. Across all specifications, we observe a negative and statistically significant effect of approximately 1%, which is consistent with a 54% reduction in abandonment in a context where the proportion is approximately 2 – 3% over the period. Our empirical findings suggest that the 4% decline in births following the abolition of the *ruota* predominantly reflects reduced marital fertility, as the decrease in legitimate births shows.²⁹ The fertility decline we observe suggests that the fertility transition in Italy may have started (or at least had its seeds) earlier than previously thought. However, more research is needed to confirm this claim and further explore the factors influencing fertility decline in nineteenth-century Italy, shedding light on its geographical heterogeneity.

The third finding is the reduction in infant mortality. The exceptionally high mortality among foundlings likely drove the reduction in infant mortality. A child raised at home had a higher chance of survival than an abandoned child, so the decrease in the abandonment share might have benefited both those children who would have been abandoned but were now outside the system and the newly abandoned — who now had better odds of surviving in less crowded institutions. The case of the nineteenth-century Madrid Foundling home offers an interesting example of the relationship between illegitimacy and infant health. Revuelta-Eugercios (2013) finds that the health of infants abandoned by married couples was poorer than that of those born to unwed women. The closure of the *ruota*, thus the end of legitimate abandonment, might have benefited the health of these newborns. In Appendix G, we perform a counterfactual analysis to try to understand whether the reduction in infant mortality can be attributed solely to children avoiding higher mortality rates of foundling homes or if the abolition of the *ruota* also influenced mortality rates of children outside these institutions. We find that 79.4% of deaths averted thanks to the abolition of the *ruota* can be attributed to the reduction of foundlings’ mortality. The remaining reduction (20.6%) likely reflects broader changes in family attitudes triggered by the closures, which affected also children outside foundling homes. As reported in the Appendix, this represents a lower-bound estimate, thus it is likely that the weight of foundling mortality reduction in explaining the overall mortality was even lower.

Finally, we find that the closure of the *ruota* contributed to the reduction of sex selectivity in abandonment, which had historically disproportionately disadvantaged girls. In accordance with recent findings by Beltrán Tapia and Cappelli (2024), we find evidence of a relationship between infant abandonment and son preference in the whole Italian peninsula. This analysis emphasizes a need for deeper investigations into the relationship between abandonment and gender discriminatory practices, including potential geographical disparities and a finer spatial disaggregation.

The *ruota* was a path-dependent institution profoundly embedded in the country’s cultural, religious, and social milieu. Its abolition represented an exogenous shock to the cost structure of child-rearing in nineteenth-century Italy by discarding an anonymous, low-cost mechanism for relinquishing unwanted offspring. This influenced households’ economic calculations regarding child-rearing costs and fertility choices, as the *ruota* was the most prevalent form of preventive check for Italian married couples. Therefore, the effects of the closure seem to be consistent with the theoretical predictions of the quality–quantity model, according to which an increase in the cost of child quantity would result in a decline in fertility. Indeed, we observed a change that altered the incentives of couples. However, this change does not align with the typical economic explanations commonly used in NHE models to justify the onset of the fertility transition. These explanations include industrialization, urbanization, the increase in waged employment for women, and the return on investments in human capital. Guinnane (2011) has challenged some of the model’s predictions, highlighting that most of the costs deemed relevant for the fertility transition did not happen at the expected time. By contrast, unlike most of the cost increases suggested by NHE models, we witness a change in the direct costs of children that did happen at the relevant time, leading to a decline in fertility. To interpret this finding, we can use the RWA (“Ready, Willing, Able”) framework developed by demographers to explain the preconditions for fertility decline (Coale, 1973; Hacker, 2016). The term *ready* is associated with economic incentives, whereas *willing* is linked to the evolution of social and cultural norms, and this might be what we are seeing here. In the case under scrutiny, it is not modern economic growth that directly generates the economic incentives. This research underscores the importance of understanding how institutional changes (i.e., the abolition of a path-dependent institution such as the *ruota*) have affected fertility decline, whereas modern economic growth alone may not be sufficient to explain it. In other words, in the spirit of Lesthaeghe and Vanderhoeft (2001), it is “culture and economics”, rather than “culture versus economics”.

By underlining the importance of being *willing*, that is of religious, political, and institutional factors in explaining the fertility decline, we aim to bridge the gap between the Q–Q predictions and the feminist economics critique. The *ruota* was a complex institution deeply embedded in Italy’s socioeconomic and cultural context. The system’s effectiveness stemmed not just from its role as a cost-minimizing option but from its legitimization through cultural and religious institutions that made it a socially acceptable

²⁹ However, the abolition does not seem to have led to an increase in the number of marriages (Appendix F). A potential explanation might reside in the fact that over time the closure resulted in a less-stigmatizing attitude towards out-of-wedlock births, thus reducing the incentive to marry.

— albeit often stigmatized — solution to unwanted pregnancies. Viewing the *ruota* solely as a means for distressed families to anonymously abandon infants overlooks its role as a surveillance mechanism for unwed mothers, concealing gender power dynamics that standard economic models fail to capture. This dual nature of the *ruota* — as both an economic mechanism and a social institution — suggests that understanding historical demographic transitions requires considering how economic incentives operate within, and are transformed by, specific cultural and institutional contexts. Its closure did not only decrease the cost of child rearing, but it was accompanied by a progressively less stigmatizing view of unwed women. This trend spread across Northern Italian provinces as it emerged from the proliferation of national and international conferences on the issues around child rearing.³⁰ Ultimately, authorities kept foundling homes open but no longer compelled unwed mothers to surrender their children. This period also saw the opening of the first public infant care centers in Northern cities, aimed at working mothers near factories (Kertzer, 1993). However, despite these improvements in the conditions of unwed mothers, the situation was still very far from women’s empowerment. This resonates with today’s academic and policy arguments concerning baby hatches in the present era. Despite the historical and contemporary distinction between baby hatches and foundling wheels, the policies surrounding their abolition and support exhibit a notable parallel. In the present context, baby hatches (or safe haven laws) can serve as a protective mechanism for infants. However, their prevalence highlights the shortcomings of the system in providing comprehensive support to vulnerable women, in investing in welfare programs, and tackling the socio-economic problems leading to child abandonment (Oaks, 2015). Nineteenth-century *ante literam* feminists like Alessandrina Ravizza, Anna Maria Mozzoni, Jesse White Mario, and Lidia Poet stressed that the abolition of the *ruota* should have been accompanied by a law enforcing the determination of paternity. But Italian law continued to protect Italian men from being held accountable.

CRediT authorship contribution statement

Giuliana Freschi: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Marco Molteni:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization.

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Appendix A. Dataset and variables

See Table 8.

Table 8	
Variable description.	
Variable	Description
Births (total, log):	Number of new births by year; births are disaggregated by gender and legitimacy status. Data collected from the <i>Movimento dello Stato Civile</i> .
Illegitimate births (total, log):	Number of illegitimate births by year, disaggregated by gender. Data collected from the <i>Movimento dello Stato Civile</i> .
Exposed births (total, log):	Number of “esposti” by year, disaggregated by gender. They differed from the previous category because their legitimacy status is unknown. Data collected from the <i>Movimento dello Stato Civile</i> .
Stillbirths (total, log):	Number of stillbirths by year, disaggregated by gender. Data collected from the <i>Movimento dello Stato Civile</i> .
Deaths (total, log):	Number of deaths by year. The data are also disaggregated by age groups. In the specifications where we look at infant mortality, we consider only deaths of people older than 1 year old. Data collected from the <i>Movimento dello Stato Civile</i> .
Infant mortality (total, log):	Number of infants who died in the first year of life, disaggregated by gender. Data collected from the <i>Movimento dello Stato Civile</i> .

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³⁰ See Ipsen (2006) for a detailed account of the policies and attitudes towards children, and in particular marginalized children in Liberal Italy.

Table 8 (continued).

Variable	Description
Gender abandoned ratio:	It is measured as the ratio between the total number of male and female foundlings multiplied by 100.
Marriages(log) :	Number of marriage acts by year from the <i>Movimento dello Stato Civile</i> .
Total expenses (province):	Yearly expenses of the provincial authority as reported in the <i>Bilanci comunali e provinciali</i> .
Total revenues (province):	Yearly revenues of the provincial authority as reported in the <i>Bilanci comunali e provinciali</i> .
Total expenses (municipality):	Yearly expenses of the municipal authority as reported in the <i>Bilanci comunali e provinciali</i> .
Total revenues (municipality):	Yearly revenues of the municipal authority as reported in the <i>Bilanci comunali e provinciali</i> .
Illiteracy of newlyweds:	Ability of spouses to sign their marriage act as reported in the <i>Movimento dello Stato Civile</i> . It is measured as a share of the total number of signed marriage acts.
Number of <i>ruote</i> :	Constructed as a categorical variable (six categories) from the report on infant abandonment published in the <i>Annali di Statistica</i> . The variable is reported in Fig. 1.
Electoral turnout:	The proportion of the electorate who exercised their right to vote is expressed as a percentage of the total number of eligible voters in 1866. Data from the <i>Annuario di Statistica</i> .
Clergy (males, females):	Number of people in the clergy per province per capita in 1871. Data come from the Italian population census of 1871.
Wet-nurses:	Number of wetnurses per capita by province from the 1871 Italian population census.
Midwives:	Number of midwives per capita by province from the 1871 Italian population census.
GDP:	Provincial estimates of GDP in 1871 provided by Chiaiese (2024).
Population:	Provincial estimates of population in 1871 provided by Chiaiese (2024).
Female literacy:	Measured as a share in 1871. Data from Ciccarelli and Weisdorf (2019).
Industry value added:	Provincial industrial value added in 1871 from Ciccarelli and Fenoaltea (2012).

Appendix B. The abolition of the *ruota*

See Table 9.

Table 9

Abolition of the *ruota* by province, 1862–1883.

Province	Year	Description	Source	Partially
Alessandria	1888	5 out of 6 <i>ruote</i> in the province were closed in the period 1884–1887. In 1887, infants were still abandoned in the <i>ruota</i> , whereas in 1888 they were not.	MAIC, Popolazione. Movimento dello Stato Civile, 1878; MAIC, Statistica delle cause di morte, 1888; Statistica della assistenza dell'infanzia abbandonata, 1894.	No
Ancona	1900	Until 1896 there were 4 <i>ruote</i> operating, whereas in 1902, they were no longer active.	Inchiesta sui brefotrofi e studi di legislazione comparata sui provvedimenti della infanzia abbandonata, 1900; MAIC, Statistica delle cause di morte, 1907.	Yes
Arezzo	1874	There were 3 <i>ruote</i> in the province, all abolished in the period 1873–75. 1874 is the year of the closure in the province's capital.	MAIC, Popolazione. Movimento dello Stato Civile, 1878.	Yes
Ascoli Piceno	1893	Year in which it was abolished in the province's capital.	Statistica della assistenza dell'infanzia abbandonata, 1894; Inchiesta sui brefotrofi e studi di legislazione comparata sui provvedimenti della infanzia abbandonata, 1900.	Yes
Avellino	post-1900		MAIC, Statistica delle cause di morte, 1907.	No
Bari	post-1900		MAIC, Statistica delle cause di morte, 1907.	No

(continued on next page)

Table 9 (continued).

Province	Year	Description	Source	Partially
Belluno	1874	There were 2 <i>ruote</i> , both abolished in the same year.	MAIC, Popolazione. Movimento dello Stato Civile, 1878.	No
Benevento	post-1900		MAIC, Statistica delle cause di morte, 1907.	Yes
Bergamo	1883	Most of the <i>ruote</i> in the province were abolished in the same year, including that in the province's capital.	MAIC, Popolazione. Movimento dello Stato Civile, 1878; Rasieri, 1884; Risultati dell'inchiesta sulle condizioni igieniche e sanitarie nei comuni del regno, 1886.	No
Bologna	1873	We take as reference the abolition in the province's capital. However, 2 municipalities kept it open up to 1894.	MAIC, Popolazione. Movimento dello Stato Civile, 1878; Inchiesta sui brefotrofi e studi di legislazione comparata sui provvedimenti per l'assistenza della infanzia abbandonata, 1900.	Yes
Brescia	1870	The 3 <i>ruote</i> in the province were closed on the same date.	MAIC, Popolazione. Movimento dello Stato Civile, 1878.	No
Cagliari	post-1900		MAIC, Statistica delle cause di morte, 1907.	No
Caltanissetta	post-1900		MAIC, Statistica delle cause di morte, 1907.	No
Campanopobasso	post-1900		MAIC, Statistica delle cause di morte, 1907.	No
Caserta	post-1900		MAIC, Statistica delle cause di morte, 1907.	Yes
Catania	1873	All the <i>ruote</i> in the province's municipalities were closed on the same date.	MAIC, Popolazione. Movimento dello Stato Civile, 1878.	No
Catanzaro	post-1900		MAIC, Statistica delle cause di morte, 1907.	Yes
Chieti	post-1900	Chieti is a specific case, because the <i>ruote</i> were closed in 36 municipalities, including the province's capital, by 1874. However, they kept operating in over 90 municipalities and for this reason we do not use the usual criteria of closure in the province's capital.	MAIC, Popolazione. Movimento dello Stato Civile, 1878; MAIC, Statistica delle cause di morte, 1907.	Yes
Como	1869	The 2 <i>ruote</i> were abolished in the same year.	MAIC, Popolazione. Movimento dello Stato Civile, 1878.	No
Cosenza	1873	According to the first enquiry, all the <i>ruote</i> in Cosenza were abolished in the same year. However, a later source reveals that in two districts, they were still open and operating by the end of the 1880s.	MAIC, Popolazione. Movimento dello Stato Civile, 1878; MAIC, Statistica delle cause di morte, 1907.	Yes
Cremona	1873		MAIC, Popolazione. Movimento dello Stato Civile, 1878; MAIC, Statistica delle cause di morte, 1894.	No
Cuneo	1868	Sources are not coherent. The first national enquiry reports that the 4 <i>ruote</i> had been closed by 1870. In a different source, we read that in the province's capital it was abolished in 1871. However, the municipalities of Salluzzo and Mondovi were together much more densely populated than Cuneo and received a higher number of abandoned infants. We follow the first national enquiry.	MAIC, Popolazione. Movimento dello Stato Civile, 1878; MAIC, Statistica delle cause di morte, 1888.	No
Ferrara	1867		MAIC, Popolazione. Movimento dello Stato Civile, 1878.	No
Firenze	1876	The <i>ruote</i> in the province were all abolished in the same year.	MAIC, Popolazione. Movimento dello Stato Civile, 1878.	No
Foggia	post-1900		MAIC, Statistica delle cause di morte, 1907.	Yes
Forlì	1893	From the enquiry on founding home, we read that in the period 1890–92, 331 infants had been abandoned through the <i>ruota</i> , whereas in 1894 none.	Statistica della assistenza dell'infanzia abbandonata, 1894; Inchiesta si brefotrofi e studi di legislazione comparata sui provvedimenti della infanzia abbandonata, 1900.	No

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Table 9 (continued).

Province	Year	Description	Source	Partially
Genova	1875	The <i>ruote</i> in the province were all abolished in the same year.	MAIC, Popolazione. Movimento dello Stato Civile, 1878.	No
Girgenti	post-1900	The <i>ruote</i> in the province were all abolished in the same year.	MAIC, Statistica delle cause di morte, 1907.	No
Grosseto	1874		MAIC, Popolazione. Movimento dello Stato Civile, 1878.	No
Lecce	post-1900	In Portoferraio, on the Island of Elba, it had been abolished in 1875, however on the mainland it remained open	MAIC, Statistica delle cause di morte, 1907.	No
Livorno	1878		MAIC, Popolazione. Movimento dello Stato Civile, 1878.	No
Lucca	1894	In 2 municipalities, it was still open in 1894: one was still active in 1893, but no longer in 1894	Inchiesta sui brefotrofi e studi di legislazione comparata sui provvedimenti per l'assistenza della infanzia abbandonata.	No
Macerata	post-1900	We use as a reference the year of the abolition in the province's capital. In 2 less-populous municipalities, they had been abolished later (but also in these two cases, the information is not consistent across different reports).	MAIC, Statistica delle cause di morte, 1907.	Yes.
Mantova	1876		MAIC, Popolazione. Movimento dello Stato Civile, 1878; MAIC, Statistica delle cause di morte, 1895.	No
Massa e Carrara	1896		Inchiesta sui brefotrofi e studi di legislazione comparata sui provvedimenti per l'assistenza dell'infanzia abbandonata, 1900.	No
Messina	post-1900		MAIC, Popolazione. Movimento dello Stato Civile, 1878; MAIC, Statistica delle cause di morte, 1895; MAIC, Statistica delle cause di morte, 1907.	No
Yes. Milano	1869		MAIC, Popolazione. Movimento dello Stato Civile, 1878.	No
Modena	1873		MAIC, Popolazione. Movimento dello Stato Civile, 1878.	No
Napoli	1875	In the province there were many <i>ruote</i> , but we consider as a benchmark the closure in Naples. The province's capital hosted a large foundling home, thus playing a central role in infant abandonment not only in the province, but also of the neighboring areas.	MAIC, Popolazione. Movimento dello Stato Civile, 1878.	No
Novara	1870		MAIC, Popolazione. Movimento dello Stato Civile, 1878.	No
Padova	1884	There were several <i>ruote</i> in the province, but we consider the closure in the province's capital.	MAIC, Popolazione. Movimento dello Stato Civile, 1878; Risultati dell'Inchiesta sulle condizioni igieniche e sanitarie nei comuni del regno, 1886; Statistica della assistenza dell'infanzia abbandonata, 1894.	Yes
Palermo	post-1900		MAIC, Statistica delle cause di morte, 1907.	No
Parma	1872		MAIC, Popolazione. Movimento dello Stato Civile, 1878.	No
Pavia	1876		MAIC, Popolazione. Movimento dello Stato Civile, 1878.	No
Perugia	1874		MAIC, Popolazione. Movimento dello Stato Civile, 1878; Statistiche delle Cause di Morte, 1888.	Yes
Pesaro e Urbino	1875	A foundling home existed in each district. The two province's capitals, Pesaro and Urbino, were equally populated. The abolition in Pesaro happened in 1873, whereas in Urbino in 1875. We choose Urbino as a reference, because it had a higher amount of abandoned infants.	MAIC, Popolazione. Movimento dello Stato Civile, 1878.	No

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Table 9 (continued).

Province	Year	Description	Source	Partially
Piacenza	1875		MAIC, Popolazione. Movimento dello Stato Civile, 1878.	
Pisa	1893	The first <i>ruota</i> to be abolished in the province was that of Campiglia Marittima. According to the enquiry on foundling homes, we learn that in the period 1890–92, 86 infants had been exposed.	MAIC, Popolazione. Movimento dello Stato Civile, 1878; Statistica della assistenza dell'infanzia abbandonata, 1894; Inchiesta sui brefotrofi e studi di legislazione comparata sui provvedimenti per l'assistenza della infanzia abbandonata.	No
Porto Maurizio	1885	The exact year is our guess because we could not find more detailed information. The rationale behind is that they seem to have been closed by 1884. As in that year we find that infants were still left in the <i>ruota</i> , we take the following year.	MAIC, Popolazione. Movimento dello Stato Civile, 1878; Risultati dell'Inchiesta sulle condizioni igieniche e sanitarie nei comuni del regno, 1886; MAIC, Statistica delle cause di morte, 1888.	No
Ravenna	1879	The two <i>ruote</i> in the district of Ravenna were abolished in 1879, other <i>ruote</i> in smaller districts had been abolished later, but we consider Ravenna's district as a reference.	MAIC, Statistica delle cause di morte, 1888; Statistica della assistenza dell'infanzia abbandonata; Risultati dell'inchiesta sulle condizioni igieniche e sanitarie dei comuni del regno, 1886.	Yes
Reggio Emilia	1875		MAIC, Popolazione. Movimento dello Stato Civile, 1878.	No
Roma	1872	We consider the closure of the <i>ruota</i> in the province's capital foundling home. However, the last <i>ruote</i> in the province closed in Orte, Viterbo, and Montefiascone in 1894.	MAIC, Popolazione. Movimento dello Stato Civile, 1878.	Yes
Rovigo	1876		MAIC, Popolazione. Movimento dello Stato Civile, 1878.	No
Salerno	post-1900		MAIC, Statistica delle cause di morte, 1907.	No
Sassari	post-1900		MAIC, Statistica delle cause di morte, 1907.	No
Siena	1876	7 out of 8 <i>ruote</i> in the province were closed between the last six months of 1875 and the first months of 1876, including that in the province's capital. The number of admissions through the <i>ruota</i> by the end of the period of analysis is 0.	MAIC, Popolazione. Movimento dello Stato Civile, 1878; MAIC, Statistica delle cause di morte, 1888.	No
Siracusa	post-1900		MAIC, Statistica delle cause di morte, 1907.	No
Sondrio	1877		MAIC, Popolazione. Movimento dello Stato Civile, 1878.	No
Teramo	post-1900		MAIC, Statistica delle cause di morte, 1907.	No
Torino	1870		MAIC, Popolazione. Movimento dello Stato Civile, 1878.	No
Trapani	post-1900		MAIC, Statistica delle cause di morte, 1907.	No
Treviso	1873	They were all shut down in May 1873. However from one source it appears that in the district of Oderzo, two municipalities kept the <i>ruota</i> open in the 1880s, but by then the number of admissions through the <i>ruota</i> is 0.	MAIC, Popolazione. Movimento dello Stato Civile, 1878; MAIC, Statistica delle cause di morte, 1888.	No
Udine	1874	In the districts of Udine and Spilimbergo it closed in December 1873; in Gemonia, it had closed in 1870.	MAIC, Popolazione. Movimento dello Stato Civile, 1878; Risultati dell'inchiesta sulle condizioni igieniche e sanitarie dei comuni del regno, 1886.	No
Venezia	1874	In the district of Portogruaro it was abolished later.	MAIC, Popolazione. Movimento dello Stato Civile, 1878; MAIC, Statistica delle cause di morte, 1888; MAIC, Statistica delle cause di morte, 1907.	Yes

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Table 9 (continued).

Province	Year	Description	Source	Partially
Verona	1875	In one of the three districts featuring the <i>ruota</i> , it was abolished in 1870.	MAIC, Popolazione. Movimento dello Stato Civile, 1878; MAIC, Statistica delle cause di morte, 1907.	No
Vicenza	1876		MAIC, Popolazione. Movimentodello Stato Civile, 1878.	No

Appendix C. Robustness checks

Table 10

Effect of the abolition of the *ruota* on infant abandonment: Robustness checks on outcome model.

Model Description	tot_exp_pc	lit_wed	exp_no_pc	revenues	stillbirths	weddings	no_deaths	all_deaths	Full Model
ATT	−0.778	−0.777	−0.781	−0.781	−0.805	−0.861	−0.837	−0.795	−0.782
S.E.	0.147	0.149	0.148	0.149	0.152	0.199	0.202	0.152	0.151
p-val	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
obs	1176	1060	1176	1176	1173	1176	1235	1235	1176
Outcome Model Controls:									
Log new births	yes	yes	yes	yes	yes	yes	yes	yes	yes
Log deaths (>1 y.o.)	yes	yes	yes	yes	yes	yes	no	no	yes
Municipality expenses per capita	no	yes	no	no	yes	yes	yes	yes	yes
Province expenses per capita	no	yes	no	no	yes	yes	yes	yes	yes
Total expenses per capita	yes	no	no	no	no	no	no	no	no
Illiteracy of newly weds	no	yes	no	no	no	no	no	no	no
Log municipality expenses	no	no	yes	no	no	no	no	no	no
Log province expenses	no	no	yes	no	no	no	no	no	no
Log population	no	no	yes	no	no	no	no	no	no
Municipality revenues per capita	no	no	no	yes	no	no	no	no	no
Province revenues per capita	no	no	no	yes	no	no	no	no	no
Log stillbirths	no	no	no	no	yes	no	no	no	no
Log weddings	no	no	no	no	no	yes	no	no	no
Log deaths (all)	no	no	no	no	no	no	no	yes	no
Treatment Model Controls:									
Female literacy 1871 (%)	yes	yes	yes	yes	yes	yes	yes	yes	yes
Wetnurses and midwives per capita	yes	yes	yes	yes	yes	yes	yes	yes	yes
Log GDP per capita (1871)	yes	yes	yes	yes	yes	yes	yes	yes	yes
Male Clergy per capita	yes	yes	yes	yes	yes	yes	yes	yes	yes
Dummy South	yes	yes	yes	yes	yes	yes	yes	yes	yes

Table 11

Effect of the abolition of the *ruota* on infant abandonment: Robustness checks on treatment model.

Model description	fem_clergy	treat2	cat_ruote	vot_sh	indVA	no_south	Full model
ATT	−0.880	−1.535	−0.865	−0.566	−0.623	−0.577	−0.782
S.E.	0.202	0.300	0.138	0.097	0.140	0.135	0.151
p-val	0.000	0.000	0.000	0.000	0.000	0.000	0.000
obs	1176	1185	1176	1176	1176	765	1176
Outcome Model Controls:							
Log new births	yes	yes	yes	yes	yes	yes	yes
Log deaths (>1 y.o.)	yes	yes	yes	yes	yes	yes	yes
Municipality expenses per capita	yes	yes	yes	yes	yes	yes	yes
Province expenses per capita	yes	yes	yes	yes	yes	yes	yes
Treatment Model Controls:							
Female literacy 1871 (%)	yes	yes	yes	yes	yes	yes	yes
Wetnurses and midwives per capita	yes	yes	no	yes	yes	yes	yes
Log GDP per capita (1871)	yes	yes	yes	yes	no	yes	yes
Male Clergy per capita	no	yes	yes	yes	yes	yes	yes
Dummy South	yes	yes	yes	yes	yes	yes	yes
Female Clergy per capita	yes	no	no	no	no	no	no
Alternative treatment	no	yes	no	no	no	no	no
Number of ruote	no	no	yes	no	no	no	no
Turnout 1866 elections	no	no	no	yes	no	no	no
Industry Value added	no	no	no	no	yes	no	no
Exclude Southern provinces	no	no	no	no	no	yes	no

Table 12Effect of the abolition of the *ruota* on new births: Robustness checks on outcome model.

Model description	tot_exp_pc	lit_wed	exp_no_pc	revenues	stillbirths	weddings	no_deaths	all_deaths	Full model
ATT	−0.033	−0.041	−0.037	−0.042	−0.039	−0.042	−0.039	−0.042	−0.042
S.E.	0.010	0.010	0.010	0.010	0.009	0.010	0.010	0.010	0.010
p-val	0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
obs	1176	1060	1176	1176	1173	1176	1237	1235	1176
Outcome Model Controls:									
Log deaths (>1 y.o.)	yes	yes	yes	yes	yes	yes	no	no	yes
Municipality expenses per capita	no	yes	no	no	yes	yes	yes	yes	yes
Province expenses per capita	no	yes	no	no	yes	yes	yes	yes	yes
Total expenses per capita	yes	no	no	no	no	no	no	no	no
Illiteracy of newly weds	no	yes	no	no	no	no	no	no	no
Log municipality expenses	no	no	yes	no	no	no	no	no	no
Log province expenses	no	no	yes	no	no	no	no	no	no
Log population	no	no	yes	no	no	no	no	no	no
Municipality revenues per capita	no	no	no	yes	no	no	no	no	no
Province revenues per capita	no	no	no	yes	no	no	no	no	no
Log stillbirths	no	no	no	no	yes	no	no	no	no
Log weddings	no	no	no	no	no	yes	no	no	no
Log deaths	no	no	no	no	no	no	no	yes	no
Treatment Model Controls:									
Female literacy 1871 (%)	yes	yes	yes	yes	yes	yes	yes	yes	yes
Wetnurses and midwives per capita	yes	yes	yes	yes	yes	yes	yes	yes	yes
Log GDP per capita (1871)	yes	yes	yes	yes	yes	yes	yes	yes	yes
Male Clergy per capita	yes	yes	yes	yes	yes	yes	yes	yes	yes
Dummy South	yes	yes	yes	yes	yes	yes	yes	yes	yes

Table 13Effect of the abolition of the *ruota* on new births: Robustness checks on treatment model.

Model description	fem_clergy	treat2	cat_ruote	vot_sh	indVA	no_south	Full model
ATT	−0.029	−0.032	−0.040	−0.029	−0.032	−0.027	−0.042
S.E.	0.010	0.012	0.010	0.010	0.011	0.012	0.010
p-val	0.003	0.008	0.000	0.004	0.004	0.019	0.000
obs	1176	1185	1176	1176	1176	765	1176
Outcome Model Controls:							
Log deaths (>1 y.o.)	yes	yes	yes	yes	yes	yes	yes
Municipality expenses per capita	yes	yes	yes	yes	yes	yes	yes
Province expenses per capita	yes	yes	yes	yes	yes	yes	yes
Treatment Model Controls:							
Female literacy 1871 (%)	yes	yes	yes	yes	yes	yes	yes
Wetnurses and midwives per capita	yes	yes	no	yes	yes	yes	yes
Log GDP per capita (1871)	yes	yes	yes	yes	no	yes	yes
Male Clergy per capita	no	yes	yes	yes	yes	yes	yes
Dummy South	yes	yes	yes	yes	yes	no	yes
Female Clergy per capita	yes	no	no	no	no	no	no
Alternative treatment	no	yes	no	no	no	no	no
Number of ruote	no	no	yes	no	no	no	no
Turnout 1866 elections	no	no	no	yes	no	no	no
Industry Value added	no	no	no	no	yes	no	no
Exclude Southern provinces	no	no	no	no	no	yes	no

Table 14Effect of the abolition of the *ruota* on infant mortality: Robustness checks on outcome model.

Model description	tot_exp_pc	lit_wed	exp_no_pc	revenues	stillbirths	weddings	no_births	Full model
ATT	−0.106	−0.104	−0.108	−0.107	−0.092	−0.110	−0.100	−0.106
S.E.	0.024	0.026	0.029	0.028	0.025	0.028	0.021	0.027
p-val	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
obs	1176	1060	1176	1176	1173	1176	1176	1176
Outcome Model Controls:								
Log new births	yes	yes	yes	yes	yes	yes	no	yes
Log deaths (>1 y.o.)	yes	yes	yes	yes	yes	yes	yes	yes
Municipality expenses per capita	no	yes	no	no	yes	yes	yes	yes
Province expenses per capita	no	yes	no	no	yes	yes	yes	yes
Total expenses per capita	yes	no	no	no	no	no	no	no
Illiteracy of newly weds	no	yes	no	no	no	no	no	no
Log municipality expenses	no	no	yes	no	no	no	no	no
Log province expenses	no	no	yes	no	no	no	no	no
Log population	no	no	yes	no	no	no	no	no
Municipality revenues per capita	no	no	no	yes	no	no	no	no
Province revenues per capita	no	no	no	yes	no	no	no	no
Log stillbirths	no	no	no	no	yes	no	no	no
Log weddings	no	no	no	no	no	yes	no	yes
Treatment Model Controls:								
Female literacy 1871 (%)	yes	yes	yes	yes	yes	yes	yes	yes
Wetnurses and midwives per capita	yes	yes	yes	yes	yes	yes	yes	yes
Log GDP per capita (1871)	yes	yes	yes	yes	yes	yes	yes	yes
Male Clergy per capita	yes	yes	yes	yes	yes	yes	yes	yes
Dummy South	yes	yes	yes	yes	yes	yes	yes	yes

Table 15Effect of the abolition of the *ruota* on infant mortality: Robustness checks on treatment model.

Model description	fem_clergy	treat2	cat_ruote	vot_sh	indVA	no_south	Full model
ATT	−0.070	−0.068	−0.099	−0.068	−0.071	−0.037	−0.101
S.E.	0.025	0.024	0.026	0.018	0.024	0.028	0.025
p-val	0.005	0.005	0.000	0.000	0.003	0.189	0.000
obs	1176	1185	1176	1176	1176	765	1176
Om Controls:							
Log new births	yes	yes	yes	yes	yes	yes	yes
Log deaths (>1 y.o.)	yes	yes	yes	yes	yes	yes	yes
Municipality expenses per capita	yes	yes	yes	yes	yes	yes	yes
Province expenses per capita	yes	yes	yes	yes	yes	yes	yes
Tm Controls:							
Female literacy 1871 (%)	yes	yes	yes	yes	yes	yes	yes
Wetnurses and midwives per capita	yes	yes	no	yes	yes	yes	yes
Log GDP per capita (1871)	yes	yes	yes	yes	no	yes	yes
Male Clergy per capita	no	yes	yes	yes	yes	yes	yes
Dummy South	yes	yes	yes	yes	yes	yes	yes
Female Clergy per capita	yes	no	no	no	no	no	no
Alternative treatment	no	yes	no	no	no	no	no
Number of ruote	no	no	yes	no	no	no	no
Turnout 1866 elections	no	no	no	yes	no	no	no
Industry Value added	no	no	no	no	yes	no	no
Exclude Southern provinces	no	no	no	no	no	yes	no

Table 16Effect of the abolition of the *ruota* on gender discrimination. Robustness checks on outcome model.

Model description	tot_exp_pc	lit_wed	exp_no_pc	revenues	stillbirths	no_births	Full model
ATT	11.434	13.339	13.731	13.044	18.310	13.697	13.964
S.E.	4.530	4.157	4.903	4.982	4.550	4.116	4.144
p-val	0.012	0.001	0.005	0.009	0.000	0.001	0.001
obs	1235	1147	1235	1235	1156	1331	1331
Om Controls:							
New birth sex ratio	yes	yes	yes	yes	yes	yes	yes
Foundlings/New Births	yes	yes	yes	yes	yes	yes	yes
Log new births	yes	yes	yes	yes	yes	no	yes
Municipality expenses per capita	yes	no	no	no	no	no	no
Province expenses per capita	yes	no	no	no	no	no	no
Illiteracy of newly weds	no	yes	no	no	no	no	no
Log municipality expenses	no	no	yes	no	no	no	no
Log province expenses	no	no	yes	no	no	no	no
Log population	no	no	yes	no	no	no	no
Municipality revenues per capita	no	no	no	yes	no	no	no
Province revenues per capita	no	no	no	yes	no	no	no
Log stillbirths	no	no	no	no	yes	no	no
Log weddings	no	no	no	no	no	yes	no
Tm Controls:							
Female literacy 1871 (%)	yes	yes	yes	yes	yes	yes	yes
Wetnurses and midwives per capita	yes	yes	yes	yes	yes	yes	yes
Log GDP per capita (1871)	yes	yes	yes	yes	yes	yes	yes
Male Clergy per capita	yes	yes	yes	yes	yes	yes	yes
Dummy South	yes	yes	yes	yes	yes	yes	yes

Table 17Effect of the abolition of the *ruota* on gender discrimination. Robustness checks on treatment model.

Model description	female_clergy	treat2	cat_ruote	vot_sh	indVA	no_south	Full model
ATT	11.255	14.745	16.165	9.682	13.674	10.093	13.964
S.E.	5.174	4.446	4.844	3.068	4.282	5.251	4.144
p-val	0.030	0.001	0.001	0.002	0.001	0.055	0.001
obs	1331	1342	1331	1331	1331	872	1331
Om Controls:							
New birth sex ratio	yes	yes	yes	yes	yes	yes	yes
Foundlings/New Births	yes	yes	yes	yes	yes	yes	yes
Log new births	yes	yes	yes	yes	yes	yes	yes
Tm Controls:							
Female literacy 1871 (%)	yes	yes	yes	yes	yes	yes	yes
Wetnurses and midwives per capita	yes	yes	no	yes	yes	yes	yes
Log Gdp (1871)	yes	yes	yes	yes	no	yes	yes
Log Gdp population (1871)	yes	yes	yes	yes	no	yes	yes
Male Clergy per capita	no	yes	yes	yes	yes	yes	yes
Dummy South	yes	yes	yes	yes	yes	no	yes
Female Clergy per capita	yes	no	no	no	no	no	no
Alternative treatment	no	yes	no	no	no	no	no
Number of ruote	no	no	yes	no	no	no	no
Turnout 1866 elections	no	no	no	yes	no	no	no
Industry Value added	no	no	no	no	yes	no	no
Exclude Southern provinces	no	no	no	no	no	yes	no

Appendix D. Gender discriminationSee [Figs. 10](#) and [11](#)

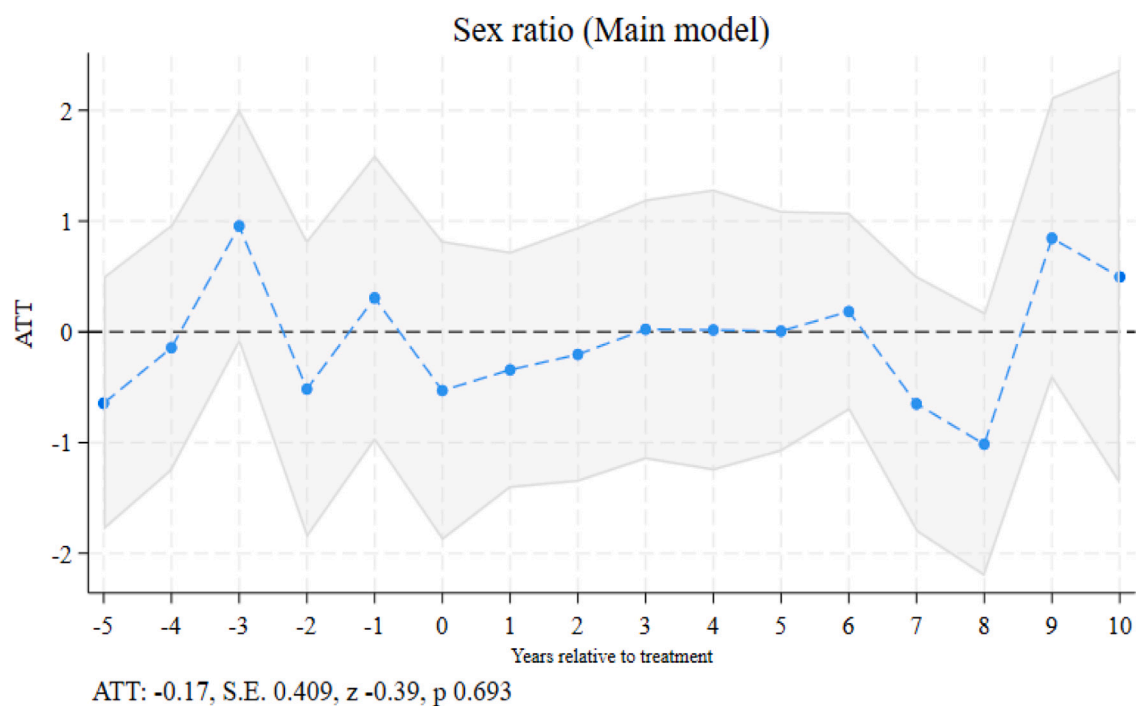


Fig. 10. Dynamic event study plot: the effect on sex ratio.

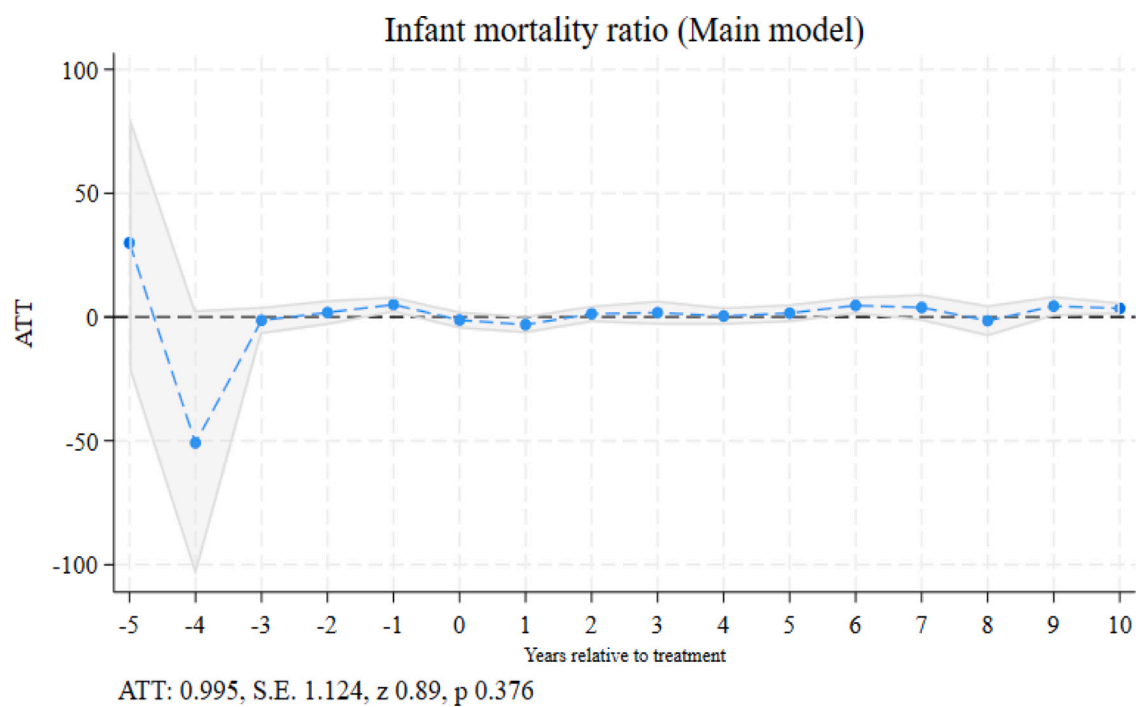


Fig. 11. Dynamic event study plot: the effect on gender ratio in infant mortality.

Appendix E. Placebo test

See Figs. 12–15.

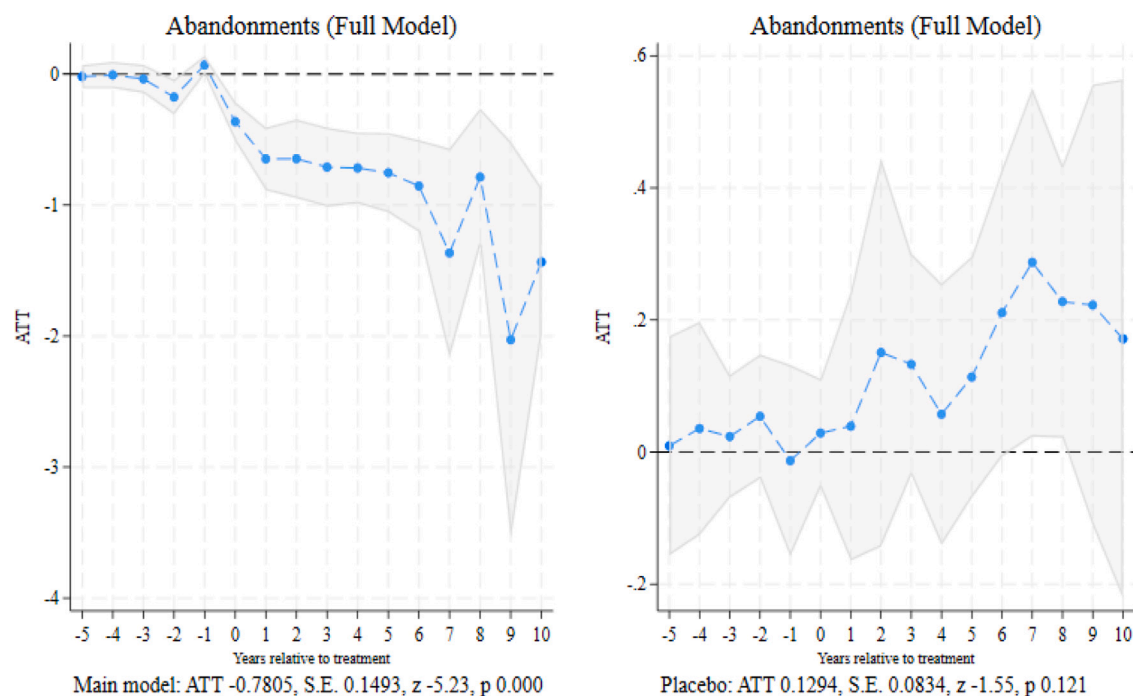


Fig. 12. Dynamic event study plot, main model vs placebo: the effect on abandonment.

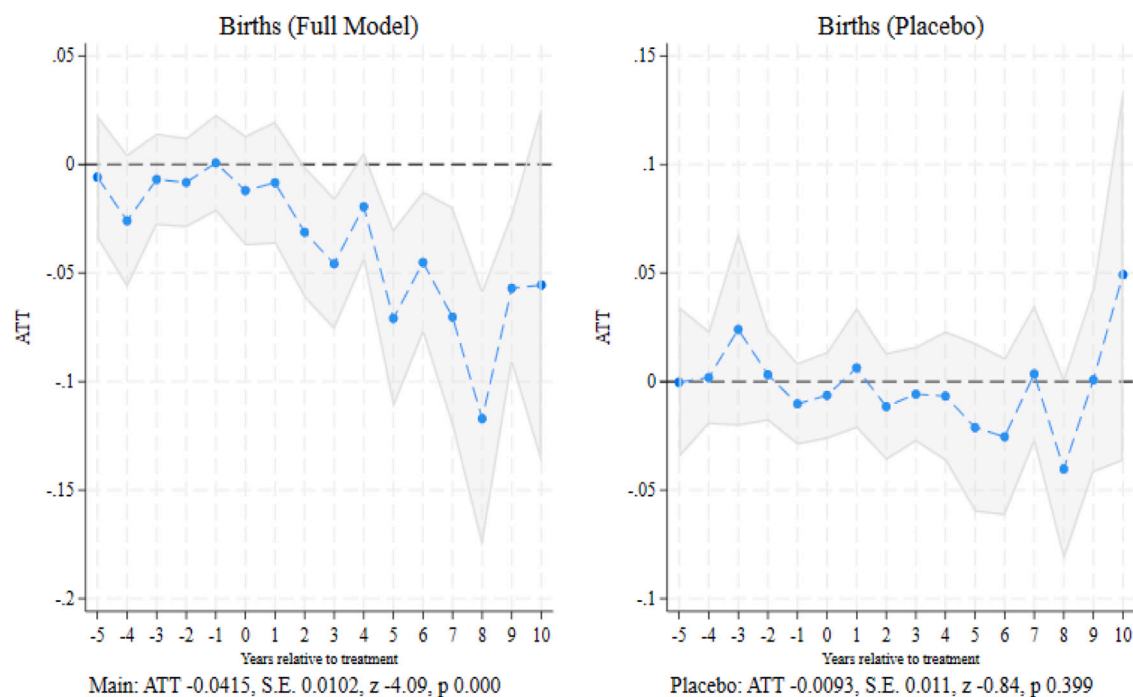


Fig. 13. Dynamic event study plot, main model vs placebo: the effect on births.

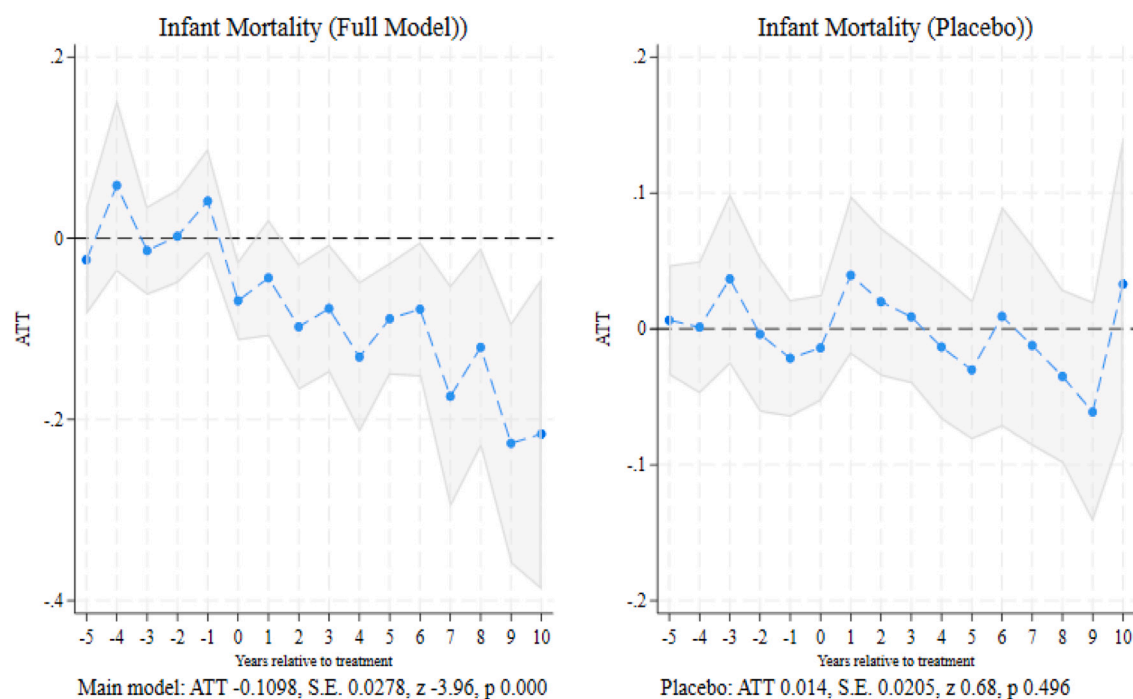


Fig. 14. Dynamic event study plot, main model vs placebo: the effect on infant mortality.



Fig. 15. Dynamic event study plot, main model vs placebo: the effect on gender discrimination in abandonment.

Appendix F. Weddings

See Table 18 and Fig. 16.

Table 18
Effect of the abolition of the *ruota* on weddings.

Model description	Twfe	Baseline	Treatment model	No partially
ATT	−0.082	−0.071	−0.039	−0.021
S.E.	0.027	0.046	0.023	0.026
p-val	0.0037	0.120	0.088	0.432
obs	1371	1360	1360	980
Treatment Model Controls:				
Female literacy (1871%)	no	no	yes	yes
Wetnurses and midwives per capita	no	no	yes	yes
Log GDP (1871)	no	no	yes	yes
Log Population (1871)	no	no	yes	yes
Male Clergy per capita	no	no	yes	yes
Dummy South	no	no	yes	no

Notes: The models present various specifications showing the effect of the abolition of the *Ruota* on the log of the number of weddings. Twfe represents the Two-Way Fixed Effects model. The models control for various covariates as specified in the Treatment Model section.

Interpretation: The ATT coefficient from the Treatment Model suggests a decrease of approximately −0.039 in the log of the number of weddings. Exponentiating this coefficient gives $e^{-0.039} \approx 0.962$, indicating that the abolition of the *Ruota* led to a decrease in the number of weddings to about 96.2% of their original value. This corresponds to a reduction of approximately 3.8%. The effect is not statistically significant.



Fig. 16. Dynamic event study plot: the effect on weddings.

Appendix G. Back-of-the-envelope calculation to estimate the contribution of avoiding foundling deaths to IMR reduction

Introduction

In this appendix, we address whether the reduction in infant mortality rate (IMR) observed can be attributed solely to children avoiding the high mortality rates of foundling hospitals or whether the abolition of the *ruota* (foundling wheel) also influenced mortality rates of children outside these institutions. Due to the dramatic mortality rates faced by abandoned children, they had a much higher probability of dying within the first year compared to non-abandoned children. The abolition of the *ruota* prevented legitimate children from being abandoned, thus mechanically reducing infant mortality as children at home had a higher chance of survival. A second channel through which infant mortality could have been reduced is through improved family care, as families had to plan fertility more carefully and invest more in their children.

Table 19
Parameters for the counterfactual analysis.

Parameter	Value per 1000	Source	Motivation
IMR_{pop,t_0}	209.6	Vecchi (2017)	Lowest value in 1865–1880
IMR_{found,t_0}	850	De Crescenzo (1873)	Highest historical value
Abandoned children	36	See text	Historical peak for 1863–1882 (1868)
Reduced Mortality Rate	101	See text	Estimated treatment effect
Reduced Abandonment Rate	549	See text	Estimated treatment effect

To explore this, we perform a back-of-the-envelope counterfactual analysis, calculating the averted infant deaths due to the abolition of the *ruota*. We determine what share of these averted deaths can be attributed to children no longer facing the high mortality rates of foundling hospitals. If this share is less than 1, the residual reduction can be attributed to fertility decisions and family care shifts. Given the data constraints, these calculations are tentative but nonetheless informative. Rather than proving that shifts in family care and fertility decisions drive the reduction in mortality, these calculations resemble a residual analysis: i.e. they show that the mechanical decrease in infant mortality observed after the abolition of the *ruota* cannot be attributed solely to the fact that children no longer died at high rates in foundling hospitals.

Our counterfactual exercise is straightforward. We calculate the averted infant deaths thanks to the abolition of the foundling wheels and then calculate what share can be explained by children no longer facing the high death rates in foundling hospitals. The intuition is that if the share is less than 1, it means that the residuals could be attributed to shifting fertility decisions in favor of fewer children but higher-quality children.

Calculation methodology

More formally, we estimate the share of averted foundling deaths over averted infant deaths (Eq. (2)). Averted foundling deaths are calculated as the difference between deaths due to abandonment before (at time 0) and after the abolition (at time 1), while averted infant deaths are calculated as the difference between the IMR before and after the abolition (Eq. (3)). Eq. (4) is the final step of the calculation. The remaining of the Appendix describes how to get to Eq. (4). To calculate the share, we use five parameters, displayed in Table 19:

$$\text{Share} = \frac{\text{Averted Foundling Deaths}}{\text{Averted Infant Deaths}} \quad (2)$$

$$= \frac{\text{Deaths Due to Abandonment}_{t_0} - \text{Deaths Due to Abandonment}_{t_1}}{IMR_{pop,t_0} - IMR_{pop,t_1}} \quad (3)$$

$$= \frac{\text{Abandoned}_{t_0} \times IMR_{found} - \text{Abandoned}_{t_0} \times (1 - \text{Reduced Abandonment Rate}) \times IMR_{found}}{IMR_{pop,t_0} - IMR_{pop,t_0} \times (1 - \text{Reduced Mortality Rate})} \quad (4)$$

We adopt conservative historical values for all parameters to ensure robustness in our back-of-the-envelope estimates and to avoid overstating the role of shifts in family planning. Each parameter and its implications are discussed below:

- **IMR_{pop,t_0} (Initial population IMR):** This parameter represents the infant mortality rate in the general population before the abolition of the *ruota*. Ceteris paribus, a lower value introduces a bias against the claim that there is a shift in fertility decisions. We take as a reference the IMR in 1880, 209.6 per 1000 live births (Vecchi, 2017). We choose this value, which is lower than the one at the beginning of the period due to the decreasing IMR in Italy at the end of the nineteenth century, because it provides a more conservative estimate.
- **IMR_{found} (Foundling mortality rate):** This parameter reflects the extremely high mortality rates in foundling institutions. A higher IMR_{found} amplifies the averted foundling deaths, strengthening the case for the mechanical impact of the abolition. A lower value diminishes this impact and increases the share of the reduction potentially attributed to family planning. We use 850 per 1000 live births, data reported by De Crescenzo (1873) relative to Naples' Foundlings hospital in 1871, to reflect a historically accurate but conservative estimate.
- **Abandoned children (Proportion of children abandoned):** This parameter determines the baseline size of the population affected by foundling mortality. Ceteris paribus, a higher value introduces a bias against the claim that there is a shift in fertility decisions. We use the historical peak for 1863–1882 of 36 per 1000 live births in 1868 to reflect a realistic but conservative estimate.
- **Reduced mortality rate:** This parameter reflects the decline in infant mortality post-abolition. We assume a reduction of 10.1%, which aligns with our empirical findings.
- **Reduced abandonment rate:** This parameter reflects the decline in abandonment post-abolition. We assume a reduction of 54.9%, which aligns with our empirical findings.

By choosing conservative values for each parameter, our estimates favor the hypothesis that most of the reduction in IMR is due to the abolition of the *ruota* and its direct effects, rather than shifts in fertility behavior or child investment.

The following equations outline the methodology for calculating averted infant deaths and their share attributed to foundling mortality.

1. Averted infant deaths in the population.

$$\text{Averted infant deaths} = \text{IMR}_{\text{pop},t_0} - \text{IMR}_{\text{pop},t_1} \quad (5)$$

Note: This formula calculates the reduction in infant mortality in the general population after the abolition of the *ruota*.

$$\text{IMR}_{\text{pop},t_1} = \text{IMR}_{\text{pop},t_0} \times (1 - \text{Reduced mortality rate}) \quad (6)$$

Note: The infant mortality rate after the *ruota*'s abolition is computed as the initial rate multiplied by the complement of the reduced mortality rate.

2. Averted foundling deaths.

$$\text{Abandoned}_{t_1} = \text{Abandoned}_{t_0} \times (1 - \text{Reduced abandonment rate}) \quad (7)$$

Note: This calculates the proportion of children still abandoned after the *ruota*'s abolition.

$$\text{Deaths due to abandonment}_{t_0} = \text{Abandoned}_{t_0} \times \text{IMR}_{\text{found}} \quad (8)$$

Note: This estimates infant deaths due to abandonment before the abolition, using the proportion abandoned and the foundling mortality rate.

$$\text{Deaths due to abandonment}_{t_1} = \text{Abandoned}_{t_1} \times \text{IMR}_{\text{found}} \quad (9)$$

Note: This calculates infant deaths due to abandonment after the abolition.

$$\text{Averted foundling deaths} = \text{Deaths due to abandonment}_{t_0} - \text{Deaths due to abandonment}_{t_1} \quad (10)$$

Note: This computes the reduction in foundling-related deaths due to the *ruota*'s abolition.

3. Share of averted deaths due to foundlings.

$$\text{Share of averted deaths due to foundlings} = \frac{\text{Averted foundling deaths}}{\text{Averted infant deaths}} \quad (11)$$

Note: This measures the proportion of averted infant deaths attributed to reduced foundling mortality. It is calculated as follows:

$$\frac{\text{Deaths due to abandonment}_{t_0} - \text{Deaths due to abandonment}_{t_1}}{\text{IMR}_{\text{pop},t_0} - \text{IMR}_{\text{pop},t_1}} \quad (12)$$

$$\frac{\text{Abandoned}_{t_0} \times \text{IMR}_{\text{found}} - \text{Abandoned}_{t_1} \times (1 - \text{Reduced abandonment rate}) \times \text{IMR}_{\text{found}}}{\text{IMR}_{\text{pop},t_0} - \text{IMR}_{\text{pop},t_0} \times (1 - \text{reduced mortality rate})} \quad (13)$$

Conclusion

Table 20 shows the results of the back-of-the-envelope calculations. The analysis reveals that 79.4% of averted infant deaths can be attributed to reduced foundling mortality. The remaining reduction (20.6%) likely stems from broader changes in family planning and child investment. It is important to stress that these are conservative estimates. For example, this model assumes that the foundlings' IMR remains constant after abolition. This assumption is most likely inaccurate, as less crowded foundling hospitals had more resources to spend per child, and thus, mortality in foundling hospitals after the abolishment would most likely decrease. However, given our calculations, if we relax this assumption, the effect would be to further reduce the share of averted deaths due to foundlings. We, therefore, prefer to maintain a conservative stance and show that even with the most conservative choice of parameters, we cannot rule out that part of the decreased IMR following the abolition of the *ruota* is due to a shift in conscious fertility planning associated with improved family care (see Table 20).

Table 20
Intermediate variables and final share value.

Back-of-the-envelope results	Value per 1000 births
Post-abolition IMR (Population)	188.4304
Post-abolition abandoned children	16.236
Deaths due to abandonment (t_0)	30.6
Deaths due to abandonment (t_1)	13.8006
Averted foundling deaths	16.7994
Averted infant deaths	21.1696
Share of averted deaths due to foundlings	79.4%

Appendix H. Share of *esposti* out of total births

See Fig. 17 and Table 21

Table 21

Model specifications and results.

Model description	Twfe	Baseline	Treatment model	Partial Treat	Not-yet only	Full model
ATT	−0.011	−0.013	−0.013	−0.011	−0.009	−0.014
S.E.	0.003	0.003	0.003	0.003	0.002	0.003
p-val	0.000	0.000	0.000	0.000	0.000	0.000
obs	1185	1331	1331	843	607	1176
Om Controls:						
Log deaths (>1 y.o.)	yes	no	no	yes	yes	yes
Municipality expenses per capita	yes	no	no	yes	yes	yes
Province expenses per capita	yes	no	no	yes	yes	yes
Tm Controls:						
Female literacy (1871%)	no	no	yes	yes	yes	yes
Wetnurses and midwives per capita	no	no	yes	yes	yes	yes
Log GDP (1871)	no	no	yes	yes	yes	yes
Log Population (1871)	no	no	yes	yes	yes	yes
Male Clergy per capita	no	no	yes	yes	yes	yes
Dummy South	no	no	yes	no	no	yes

Notes: The treatment is associated with a 1.4 percentage point reduction in the share of abandoned children over new births on average among the treated units. Considering that between 2 and 3% of new born children were abandoned — depending on the year and province — this is a large effect.

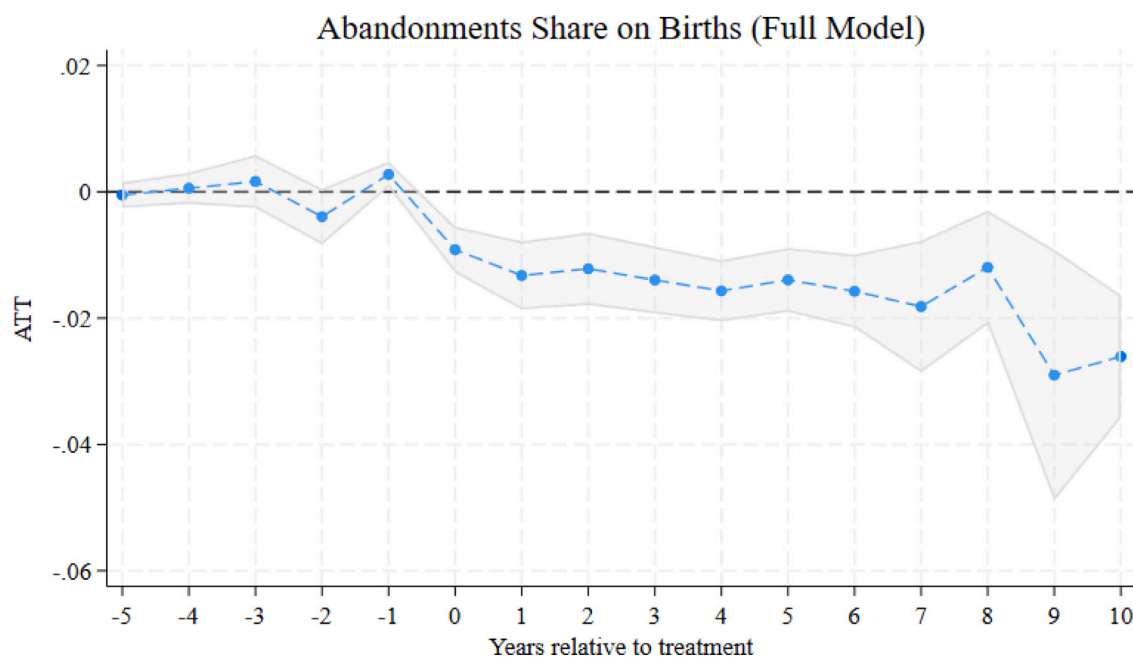


Fig. 17. Dynamic event study plot: the effect on infant abandonment (share on births).

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Appendix I. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.eeh.2025.101681>.

Data availability

Data will be made available on request.

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